

Operating and Maintenance Instruction

Customer..... Loesche GmbH
 Order no. of customer 45018650
 Project..... NARAYANGONJ
 Loesche Code no. MC1M10.010.43.00.01
 MC1M10.010.43.00.02
 Commission no.. 485060-01

Relating printscripts and data sheets

• Installation and Servicing of central lubricating systems	Printscript	X 01 BA 1
• Lubrication pump FZ-A / B	Printscript	21112 BA 2
• Motorinformation	Data sheet	76911-5133
• Ultrasonic level switch	Printscript	76951-6011-1
• Pressure relief valve NU-A	Printscript	38132 PB 1
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Spare parts drawing and spare parts list	Printscript	35541 E 2
• Monitoring switch	Printscript	66925 P 1

Installation and Servicing of Central Lubricating Systems

Operating instruction

1. Pump

- 1.01 Install the pump where it is easily accessible, in a place free from dirt. Pay attention to the operating temperature range (dependent on the type of pump and lubricant).
- 1.02 Install it at a central point in the system to be lubricated, so that with short lengths of piping, the pressure drop can be kept to a minimum thus ensuring sufficient pressure to overcome the back-pressure at the lubrication points.
- 1.03 The pump reservoir should be big enough to ensure infrequent replenishing.
- 1.04 Pumps with electric motors often need to rotate in a particular direction. Pay attention to the arrow on the motor or the pump casing. If the direction of rotation is not that shown by arrow, interchange two of the leads of 3-phase A.C. motors. The operating instructions for the particular pump should be followed in all cases.
- 2.11 When vehicles are used at the site, in mining or in agriculture, take special steps to ensure that piping cannot be pulled down, particularly in winter when there is ice and snow, e.g. by installing the piping in load-bearing parts of the plant, sufficiently high up and on the side furthest away from the normal direction of travel of the vehicles.
- 2.12 High temperature accelerates the aging of lubricants, low temperature increases the resistance offered by the piping and, in certain circumstances, can lead to the lubrication being inadequate, or to damage to the pump; piping should therefore be installed so that, as far as possible, it is not exposed to very high or very low temperatures.
- 2.13 For mobile lubrication points of difficult access, use rubberized fabric hoses or nylon pipes, or, if heavily loaded, heavy steel-braided hoses.

2. Pipes and hoses

- 2.01 The use of seamless steel pipes to DIN 2391 is recommended. For systems operating in severe conditions, it is recommended to use copper pipes, externally tinned steel pipes or stainless steel pipes.
- 2.02 Use a pipe cutter, and not a saw, for cutting pipes, so that no swarf is produced.
- 2.03 Always bend in the cold state using a bending device, to avoid scale.
- 2.04 Do not use sharp pipe bends.
- 2.05 Thoroughly clean the pipes, dislodge and remove all loose matter, and blow through; fill the pipes with lubricant as soon as they are installed, if possible.
- 2.06 Remove any rust or mill scale present inside, if necessary pickle and then neutralize.
- 2.07 Cover the openings of pipes which have been cleaned, for transport.
- 2.08 In oil central lubrication systems, install the piping with a slight slope upwards to the end of the line, to avoid air locks. If this is not possible, arrange the distributor so that the system can be bled via the distributor.
- 2.09 Avoid syphons - so-called sags - for instance when you have to bypass obstacles. In such syphons, air inclusions accumulate which cannot be eliminated during venting.
- 2.10 Fix the pipes to the machine to be lubricated or to a wall, where they can be seen and are easily accessible, using enough pipe clips to protect them from mechanical damage.

3. Pipe connections

- 3.01 For pipes of outside diameters up to about 42 mm, use non-soldered pipe unions for preference.
- 3.02 Make pipe joints in accordance with the supplier's instruction (see page 4).
- 3.03 Do not use hemp; use sealing tape or gaskets.

4. Filter

- 4.01 In some pump types, the filter is integrated in the pressure connection of the pump.
- 4.02 External filtration
- 4.03 At intervals depending on the amount of dirt, unscrew and remove the filter, clean it in paraffin and blow through with compressed air.

5. Distributor

- 5.01 Fix the distributor to the machine being lubricated, or on a wall, in such a way that it can not be pulled off by normal forces acting on it.
- 5.02 If possible, arrange the distributor so that it can be easily bled and/or replaced. Installation in a branch line is preferable to an installation in the main delivery line itself.
- 5.03 Take care that the distributors are not twisted when screwing together distributor segments (e.g. progressive distributors) because this might result in the fact that the pistons get stuck.
- 5.04 In severe conditions it is recommended to install distributors with the adjustment and indicating devices facing downwards.

6. Switching units at the end of the line

- 6.01 Switching units are often used at the end of the line in single-line and dual-line systems, to signal a maximum or minimum pressure or differential pressure. Since such units are liable to gum up after a time when grease is used, a distributor should be installed downstream from the switching unit.
Secondary delivery lines leading to the pressure switch are filled with oil appropriately.

7. Pressure relief valve

- 7.01 A pressure relief valve should always be fitted in the delivery line of a pump driven by an electric motor, so that the pump is not damaged if, while it is working, a distributor becomes jammed or a lubrication point stopped up, or the main delivery line is inadvertently closed.
- 7.02 The pressure release line from the relief valve should be connected to the suction side of the pump or to the reservoir.
- 7.03 The pressure relief valve should not be set at a higher pressure than is permissible for the pump.
(Some pumps have a built-in pressure relief valve).

8. Burst discs

- 8.01 Some pumps (e.g. BS-B), in addition to pressure relief valve, also have built-in burst discs, which burst and thereby protect the pump if the pressure rises too high (pressure relief valve wrongly set).

9. Pressure gauge

- 9.01 A pressure gauge for checking the pressure should always be installed on the pump delivery side, in the main delivery line.
- 9.02 In single-line systems for grease, a pressure gauge should also be fitted at the end of the line.
- 9.03 In dual-line systems, a pressure gauge should be fitted at the end of each of the two main delivery lines.
- 9.04 The secondary delivery lines leading to the pressure gauges are filled with oil appropriately as grease might suffer from ageing after a longer period of time.

10. Commissioning of the system

- 10.01 Avoid removing the cover to fill up the reservoir on the pump; instead, use a filling pump and fill the reservoir through the filling connection.
- 10.02 If the filling line is not permanently installed, a quick release coupling should be used.
- 10.03 When filling, take care that the reservoir does not overflow.
- 10.04 It is preferable to fit an electrical level indicator to switch off the filling pump when the level reaches the permitted maximum.
- 10.05 For the initial filling of pumps and lubricant reservoirs with grease, it is advisable, with most pumps, to put oil in first; this fills all the cavities and prevents air locks which could lead to the pump failing to develop sufficient pressure when it is started up. If the bearings are of a type which should not be lubricated with oil, even for a short time, the pump should be operated until air-free grease issues from the ends of the lines.
- 10.06 On pumps having an anti-cavitation plate and which are filled through the top of the reservoir after the cover has been removed, withdraw the plate and after filling, press the plate down on to the grease a number of times to drive out the air.
- 10.07 Do not connect the distributors before lubricant comes bubble-free out of the feed line.
- 10.08 Do not connect the lubrication points until air-free lubricant issues from the lines.
- 10.09 Cocks for checking the supply of lubricant may be fitted, if desired, in front of the lubrication points.

11. Types of lubricants and frequency of lubrication

- 11.01 Types of lubricants and frequency of lubrication are specified by the supplier or manufacturer of the machine being lubricated. In general, when starting up, it is better to over-lubricate rather than underlubricate, but in the course of time, after adequate observation of the lubrication points, it is often possible to reduce considerably the quantity of lubricant used.

12. Electrical system

- 12.01 Connection of the power supply and changing of electrical parts if required, should be undertaken only by a qualified electrician.

13. Spares

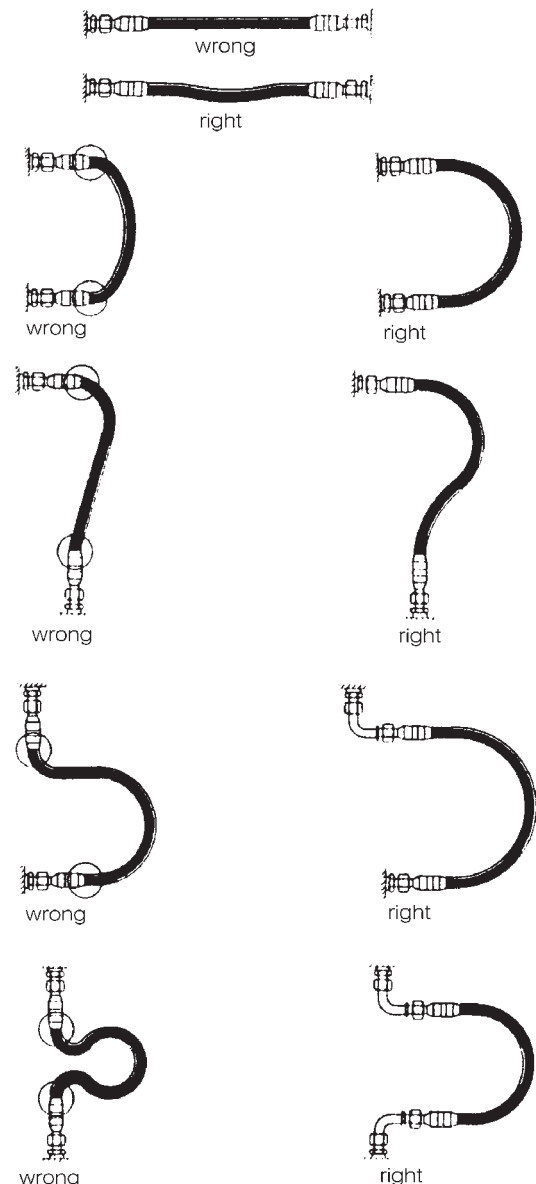
- 13.01 When ordering spares, state the following:
Designation of the pump
Code no.
Year of manufacture
Quantity, designation and Code no. of the spares required.
- 13.02 Note that in almost all pumps and distributors, the plungers have been lapped into or matched to their cylinders. These parts must not be interchanged. Plungers removed during servicing must always be replaced in the bore from which they were taken. If replacement is necessary, the two parts must be ordered together.
- 13.03 It is recommended to keep a stock of parts subject to wear, for all major units. The most important parts subject to wear are specially marked in the parts list.
- 13.04 Repair of particularly old discontinued pumps or units is sometimes uneconomical. In that case, the pump or unit should be replaced by one from our current range.
- 13.05 It is often not possible to give a firm estimate for a repair until the unit has been dismantled. If no order follows the quotation for repair, the enquirer must bear the cost of dismantling and of reassembling if required.

14. Installation of hoses

- a) Hoses should be of such a length and installed in such a way, that
1. They can be installed and removed without difficulty,
 2. The radius of curvature of the hoses is not less than the minimum stated,
 3. The hose is not kinked, either when the parts connected by it are at rest or in motion. There is a risk of this particularly at the point where the hose is connected to the fitting.

The points where the risks exist are circled in the drawings. If the space available for installation is restricted, kinking of the hose at its connection to the fitting can be avoided by the use of an elbow.

- b) Movement of the part to which the hose is connected, should be only in the plane in which the hose is installed. Rotation round an axis which is not perpendicular to the plane of installation, is not permissible.



15. Assembly of olives

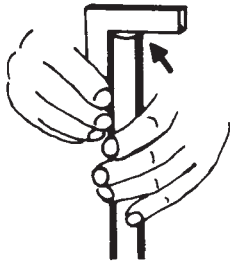


Fig.1 Cut off the end of the pipe square.

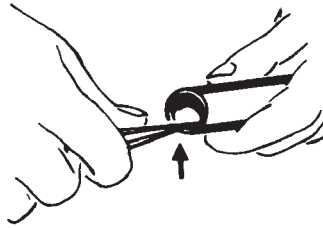


Fig. 2 Deburr the inside of the pipe.

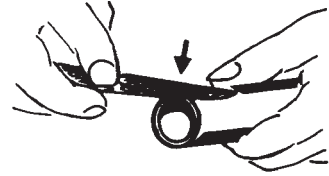


Fig. 3 Deburr the outside of the pipe.

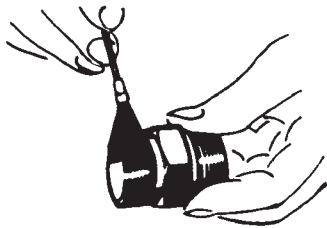


Fig.4 Thoroughly oil the thread.



Fig. 5 Oil the taper.

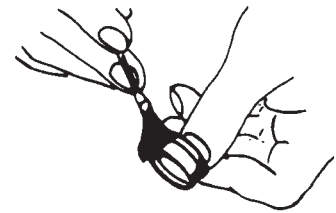


Fig. 6 Thoroughly oil the olive all over.

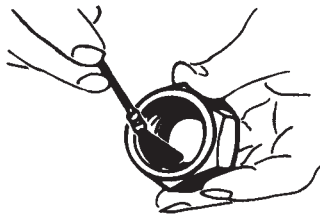


Fig.7 Thoroughly oil the thread of the union nut; do not use grease.

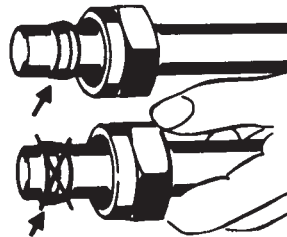


Fig. 8 Slide the olive on correctly - other side the fitting will be wrong.

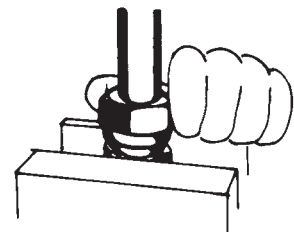


Fig. 9 Tighten the union nut by hand until it is felt to come to a stop.

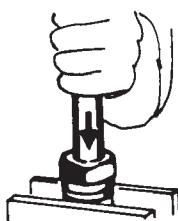


Fig.10 Press the pipe into the taper against the stop. Mark the pipe and nut.

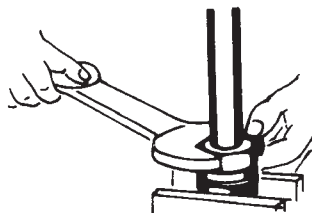


Fig. 11 Tighten the nut by about 3/4 of a turn. This causes the olive to grip the pipe. The pipe must not turn.

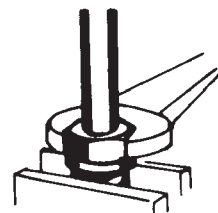


Fig. 12 Tighten finally by about 3/4 of a turn. This causes the olive to cut into the pipe and results in a visible collar over its cutting edge.

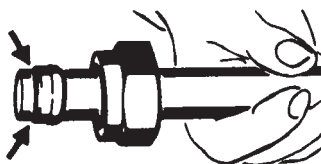


Fig.13 Unscrew the nut and check whether the visible collar takes up the space in front of the cutting edge.

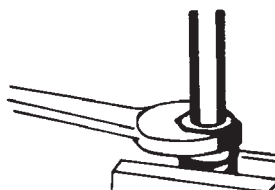


Fig. 14 Tighten the nut slightly after the connection has been completed and every time after the nut has been slackened.

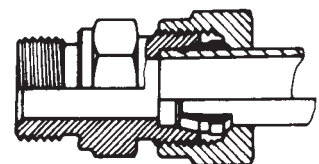


Fig. 15 How the union functions after the nut has been tightened.

Lubrication pump FZ - A and FZ - B

Operating instructions

APPLICATION

The FZ-A lubrication pump is designed as a single-plunger pump with up to 12 outlets. It is predominantly intended for direct-feed centralized lubrication systems. The pump is usually driven by the flange motor or by the machine to be lubricated via a ratched drive or a coupling or, optionally, via a transmission gear box whereby allowing for rotation between the drive shaft and the pump plunger within the range of 3 : 1 up to 2800 : 1 (for motor-driven pumps between 215 : 1 and 2880 : 1). The FZ-B type lubrication pump is similar to the FZ-A type, it has either one combined single outlet with a maximum delivery of 12 x 0.1 cm³ or with 2 outlets respectively delivering 6 times the quantity of lubricant as compared to the FZ-A.

PRINCIPLE OF OPERATING (Fig 1 and 2)

The worm driven by the machine to be lubricated with its worm wheel (1) which has two dog pins on top in the recess of which the crosspiece (2) engages. In further two grooves of the crosspiece engages the dog pin of the cam plate (3) thus allowing the cam plate and the plunger (4) to rotate. The compression spring (5) located between the crosspiece and the cam plate exercises an upward force on to the cam plate. There is a cam on top of the cam plate, which, while the cam plate rotation, actuates six adjusting spindles (6) arranged round the periphery of the pump and initiates each time the suction stroke of the plunger. The lubricant is delivered under slight pressure into the pump suction chamber (S).

On the lower part of the cam plate are six further cams which, after each revolution, actuate a pressure cam (7) inside the pump housing, each time causing the plunger to execute the compression stroke.

On pumps with up to six outlets, the plunger has three vertical grooves (N), which terminate into an annular channel (R). On pumps with up to 12 outlets the plunger has two sets of three vertical grooves each terminating into an annular channel. During the suction stroke lubricant is squeezed from the suction chamber (S) into the vertical slots of the plunger and from there into the annular channels (R).

During the compression stroke, the lubricant is forced out of the annular channel via a vertical groove into the compression channel (D).

On a continuous operation the pump is designed to overcome a backpressure of 200 bar, for a short time up to 250 bar is admissible. To prevent overpressure safety valves can be built into the pipe lines downflow.

ROTATIONAL DIRECTION OF THE DRIVE

All pumps can be used for anticlockwise or clockwise rotation (at choice) without change in performance and without modifications to the pump. Excepted are pumps being provided with a tank with a particular delivery insert on the right. For these pumps, and if it deals with free shaft end versions with anticlockwise rotation and with step-down gear with and without flange motor and with clockwise rotation, it is forbidden to change the direction of rotation.

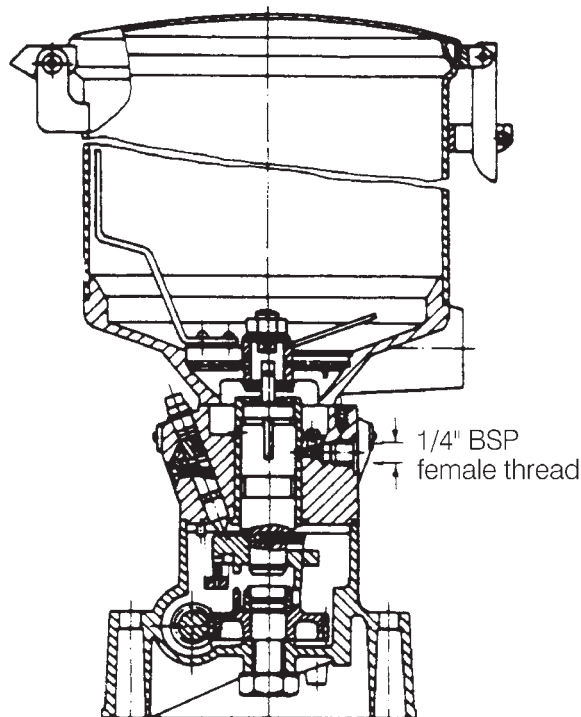


Fig. 1 section through FZ-A lubrication pump with 6 outlets

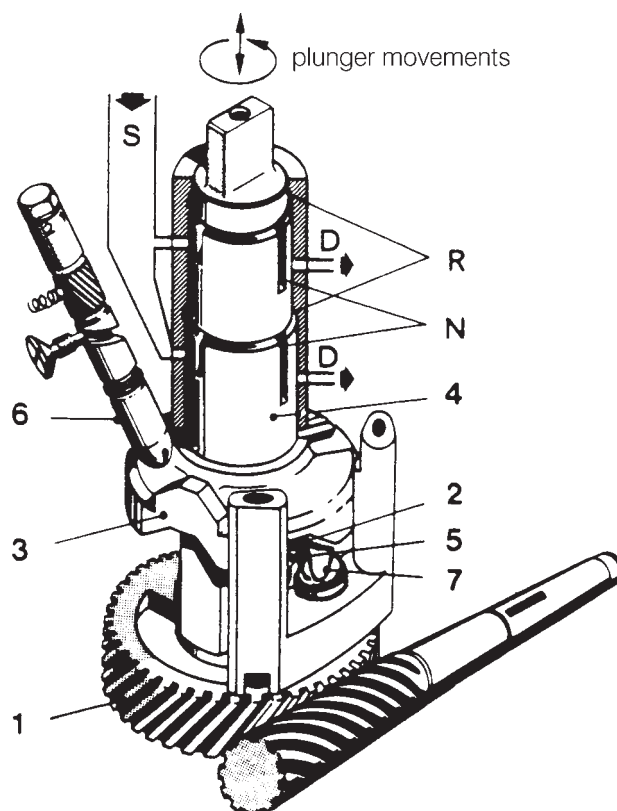
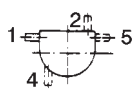
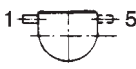


Fig. 2 principle of operation (type FZ-A with 12 outlets)

CODE NO. FOR LUBRICATION PUMP FZ - A / B

Denomination	Remarks	Code no.
with ratchet drive		21112 - 2100
rotary with free end stud		21112 - 4100
with reduction gear box		1 21112 - 6310
		2 21112 - 6320
		4 21112 - 6340
		5 21112 - 6350
with reduction gear box and integral flanged motor		1 21112 - 8210
		5 21112 - 8250

GENERAL LAYOUT (sub-assemblies)

Sud-assembly 1 _____ reservoir
 Sud-assembly 2 _____ pump body
 Sud-assembly 3 _____ drive
 Sud-assembly 4 _____ loose components groups
 Sud-assembly 5 _____ three phase AC motor
 Sud-assembly 6 _____ level switch
 Sud-assembly 7 _____ filling valve
 Sud-assembly 8 _____ solderless self-tapping pipe union
 The sub-assemblies are interchangeable.

RESERVOIR (sub-assembly 1)

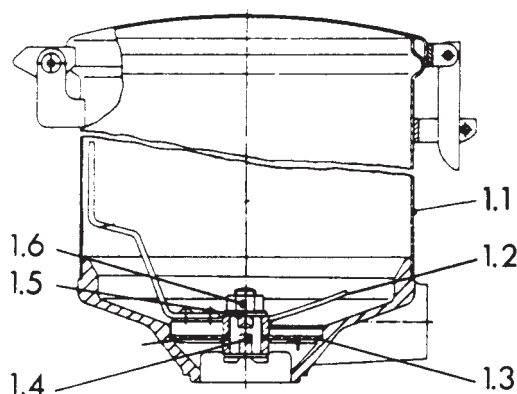


Fig 3 reservoir

The metal reservoir is available with capacities of 2.5; 8; 15 or 30 litre respectively, at choice. It is a hinge-lid-type, which can be secured with a padlock, if necessary, to prevent tampering. The reservoir contains the wedge assembly (1.1) consisting of a scraper to scrape the grease of the walls and a wedge plate to generate a slight pressure inside the suction chamber. A strainer (1.2) is fitted on the reservoir bottom to prevent impurities from penetrating into the suction chamber. A strainer (1.3) in tank bottom protects the inlet chamber of the pump from dirt.

ATTENTION !

If heavy vibrations are likely to occur at the erection place, it is essential that the version with 30 l tank is provided with a sufficient support.

Item	Pcs.	Denomination	Code no.		Rotational direction of the delivery insert
1	1	reservoir	2,5 l 63721 - 1411	•	rights and lefts
			8 l 63721 - 1451	•	
			15 l 63721 - 1402	•	
			30 l 63721 - 1461	•	
			15 l 63721 - 1441*	•	
			2,5 l 63721 - 1481	•	rights
			8 l 63721 - 1471	•	
			15 l 63721 - 1401	•	
			30 l 63721 - 1491	•	
			15 l 63721 - 1361	•	
1.1	1	reservoir (with wedge plate assy.)	2,5 l 63721 - 1311	•	rights and lefts
			8 l 63721 - 1351	•	
			15 l 63721 - 1371	•	
			30 l 63721 - 1361	•	
			15 l 63721 - 1361	•	
1.2	1	delivery insert	62631 - 1301	•	rights and lefts
			62631 - 1311	•	
			62631 - 1392	•	
			62631 - 1321	•	
			62631 - 1391	•	
			62631 - 1331	•	rights
			62631 - 1341	•	
			62631 - 1351	•	
			62631 - 1351	•	
			62631 - 1351	•	
1.3	1	strainer	73671 - 4337	•	
1.4	1	driving slot	71551 - 7216	•	
1.5	1	washer B 13 DIN 125 - St	74185 - 1094	•	
1.6	1	hexagon head nut M 12 DIN 934	74175 - 1314	•	

*) only for ratchet drive

PUMP BODY (sub-assembly 2)

The pump body comprises a control sleeve to guide the plunger and, for each outlet, an adjustable spindle (2.5) to pre-set the stroke of the plunger and consequently the quantity delivered (up to 6 outlets) or each two combined outlets one top of the other (7 up to 12 outlets).

Order of outlets see picture 4. The outlets 1 - 6 are arranged in the upper row. Outlets being placed one above each other can only be joined in the order shown on picture 4. Fusions begin at the outlets 1/7 and follow the direction of the arrow (see table on page 3). If the total grease volume is delivered via 1 outlet, any outlet can be opened (pump FZ-B: outlet 1 provided in our works). If the deals with 2 outlets each of them delivering half the total output volume, one outlet each of the upper and lower row can be opened (pumps FZ-B: outlets 1 and 7 provided in our works). The outlets, which are arranged one above the other can only be connected in the sequence shown in fig. 4, starting with 1/7 and up to 6/12. To combine several outlets start with outlets 1 - 7 in the direction of an arrow (see schedule page 3). With the grease lubrication pump FZ-B, the quantity delivered by all the outlets is taken from a single outlet or the quantity delivered by six outlets each is taken from one outlet and the quantity delivered by the other six outlets is taken from a second outlet. Leaving our works the pump FZ-B is supplied with outlets 1 or 1 and 7 open respectively.

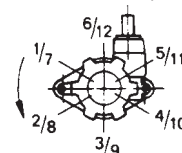


Fig. 4 denomination, position and sequence of the outlets

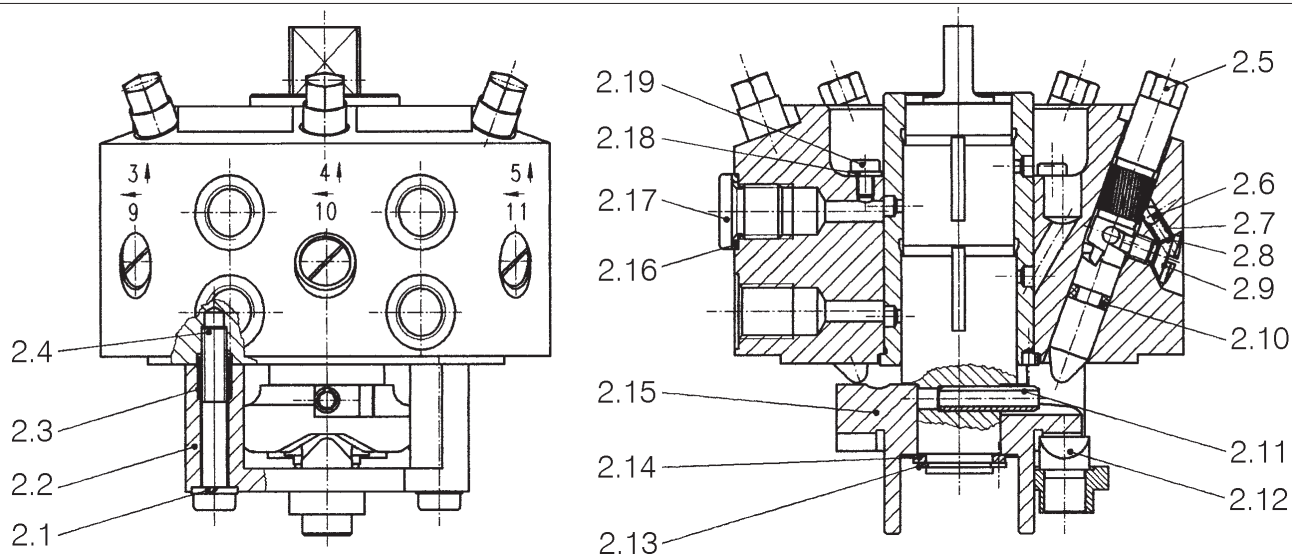


Fig. 5 pump body type FZ-A

Item	Pcs.	Denomination	Code no.	No. of outlets		Rep.
2	1	Pump body FZ - A	62227 - 2211	1	max. quantity delivered per outlet for one rotation of the plunger 0,1 cm³	I, III I, III
			62227 - 2221	2		
			62227 - 2231	3		
			62227 - 2241	4		
			62227 - 2251	5		
			62227 - 2261	6		
			62227 - 2311	7	for one rotation of the plunger 0,1 cm³	
			62227 - 2321	8		
			62227 - 2331	9		
			62227 - 2341	10		
			62227 - 2351	11		
			62227 - 2361	12		
		Pump body FZ - B	62227 - 2411	6	max. quantity delivered per outlet per rotation of the plunger	
			62227 - 2421	7		
62227 - 2431	8					
62227 - 2441	9					
62227 - 2451	10					
62227 - 2461	11	1 x 0,2 cm³ + 10 x 0,1 cm³				
Pump body FZ - B	62227 - 2521	1	1 x 1,2 cm³ (12 x 0,1 cm³)			
	62227 - 2511	2	2 x 0,6 cm³ (6 x 0,1 cm³)			
2. 1	2	Lock washer 6 DIN 7980	74185 - 3034			
2. 2	1	Bridge	72512 - 1321			
2. 3	2	Crover lock pin 8 x 10 DIN 7346	74225 - 6084			
2. 4	2	Cheese-head screw M 6 x 40 DIN 7984	74102 - 1184			
2. 5	1 - 6	Adjusting spindle	71543 - 1111			
2. 6	1 - 6	Ball 4 mm 5 DIN 5401	71961 - 4124			
2. 7	1 - 6	Pressure spring	71612 - 3112			
2. 8	1 - 6	Sealing ring A 8 x 14 x 0,5 DIN 7603 - Cu	72712 - 1723			
2. 9	1 - 6	Trunnion screw	74121 - 1111			
2.10	1 - 6	O-ring 5,3 x 2,4 NB 70	72711 - 1213			
2.11	1	Crover lock pin 6 x 24 DIN 1481	74225 - 5184			
2.12	1	Pressure cam	71542 - 1121			
2.13	1	Circlip 16 x 1 DIN 471	74186 - 1074			
2.14	1	Retaining disk 16 x 22 x 2 DIN 988	74185 - 1984			
2.15	1	Cam plate	72511 - 3431			
2.16	0 - 11	Sealing ring A 14 x 18 DIN 7603 - Cu	72712 - 1214			
2.17	0 - 11	Screw plug	74161 - 7811			
2.18	1 - 6	Sealing ring A 4 x 8 DIN 7603 - Cu	72712 - 1014			
2.19	1 - 6	Cheese-head screw M 4 x 6 DIN 912	74102 - 1994			
2.20	0 - 5	Plug	73418 - 2411			

Remarks see page 7 - repair kits

DRIVE (sub-assembly 3)

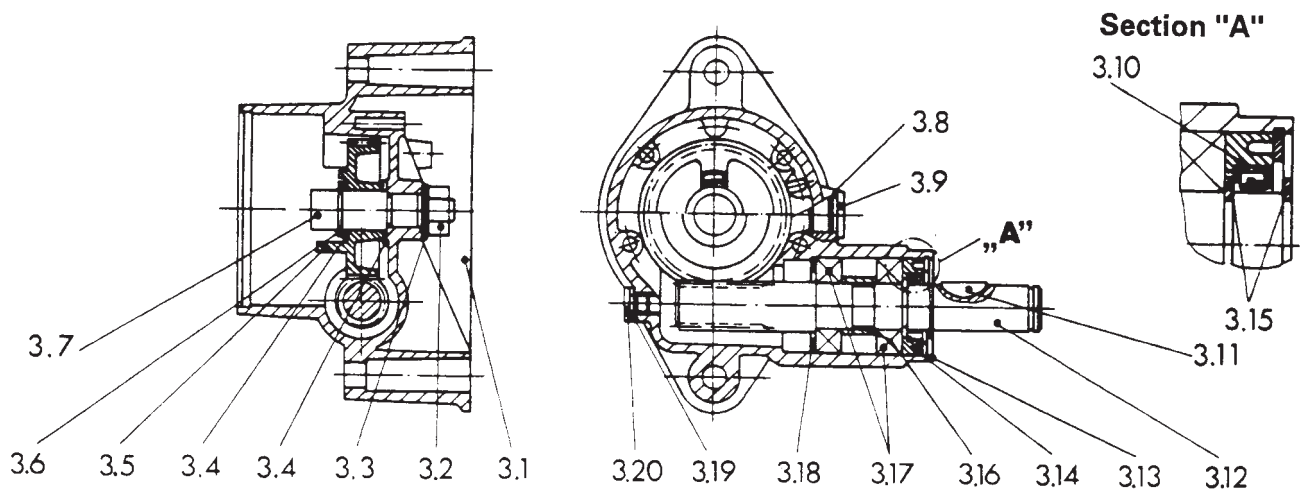


Fig. 6 rotary drive with free-end stud

Drive, rotary, with free-end stud

The sub-assembly includes the gear housing with the drive shaft (3.12) and the worm gear (3.5). The drive shaft runs in ball bearings. The plug (3.9) is removed to fill oil into the gear housing. The plug (3.20) is removed to facilitate removal of the shaft with a punch, should this become necessary. The pump may be driven clockwise or anticlockwise, at choice.

Ratched drive with oscillating lever

The sub-assembly constitutes an extension of sub-assembly with free-end shaft containing those parts necessary for the ratched drive.

The oscillating lever (3.34) is to be actuated to and for via an eccentric, crank or any other component of the machine to be lubricated thus causing the ratched drive (3.23 - 3.26) to rotate the worm gear (3.5) always in the same direction regardless of the direction of rotation of the drive shaft.

Drive with reduction gear box or with transmission gear and integral flanged motor

This assembly consists of the sub-assembly with free-end shaft plus a second worm gear box.

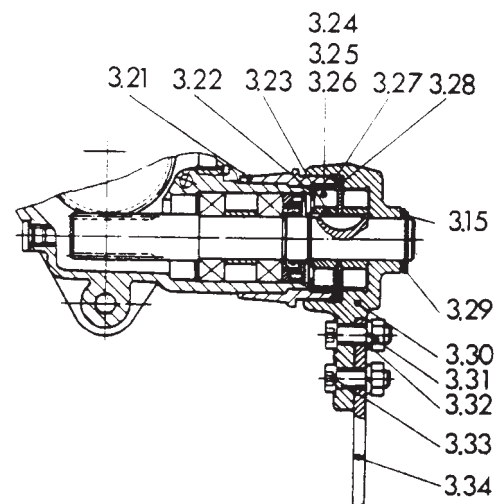


Fig. 7 ratched drive

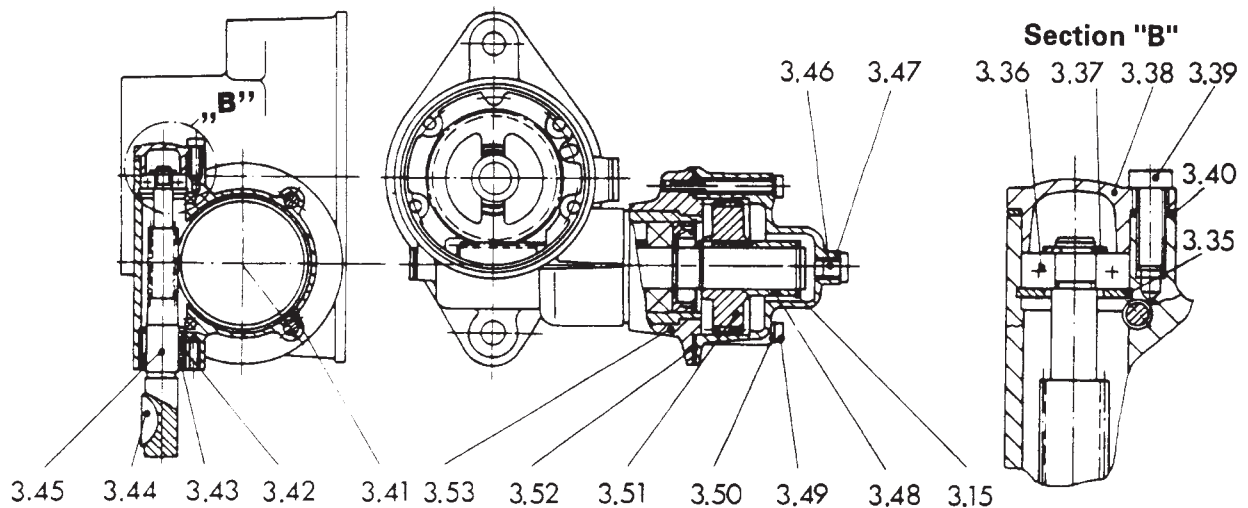


Fig. drive with reduction gear box or with transmission gear and integral flange motor

Item	Piece	Denomination	Code no.	3 : 1	12 : 1	25 : 1	50 : 1	3 : 1	12 : 1	25 : 1	50 : 1	95 : 1	215 : 1	345 : 1	710 : 1	1420 : 1	2880 : 1	Repair kits
3.	1	Rotary drive with free-end stud	61171 - 1711	•														
			61171 - 1721		•													
			61171 - 1731			•												
			61171 - 1741				•											
		Ratched drive	61161 - 1411					•										
			61161 - 1421						•									
			61161 - 1431							•								
			61161 - 1441								•							
		Rotary drive with reduction gear box or with transmission gear box and flange motor	61121 - 1411									•						
			61121 - 1471										•					
			61121 - 1421											•				
			61121 - 1431												•			
			61121 - 1441													•		
			61121 - 1461														•	
3. 1	1	Housing (not available as single part)	—															
3. 2	1	Hexagon-head nut M 12 DIN 934 - 8	74175 - 1314	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3. 3	1	Washer B 13 DIN 125 - St	74185 - 1094	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3. 4	2	Retaining disk 18 x 25 x 1 DIN 988	74185 - 2034	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3. 5	1	Worm wheel (3 : 1)	71354 - 3141	•				•				•						IV
		Worm wheel (7 : 1)	71354 - 3151										•					IV
		Worm wheel (12 : 1)	71354 - 3131		•				•						•			IV
		Worm wheel (25 : 1)	71354 - 3121			•				•						•		IV
		Worm wheel (50 : 1)	71354 - 3111				•				•						•	IV
3. 6	1	Circlip 18 x 1,2 DIN 471	74186 - 1094	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3. 7	1	Bolt	71552 - 1231	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3. 8	1	Sealing ring A 16 x 20 DIN 7603 - Cu	72712 - 1274	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3. 9	1	Screw plug M 16 x 1,5 DIN 908	74106 - 1184	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3.10	1	Radial seal A 20 x 30 DIN 6503	72712 - 2064	•	•	•	•	•	•	•	•	•	•	•	•	•	•	I , III
3.11	1	Woodruff key 6 x 9 DIN 6888	71951 - 4064	•	•	•	•	•	•	•	•	•	•	•	•	•	•	I
3.12	1	Drive shaft (3 : 1)	71354 - 1941	•				•				•						IV
		Drive shaft (7 : 1)	71354 - 1951										•					IV
		Drive shaft (12 : 1)	71354 - 1931		•				•						•			IV
		Drive shaft (25 : 1)	71354 - 1921			•				•						•		IV
		Drive shaft (50 : 1)	71354 - 1911				•				•						•	IV
3.13	1	Circlip 42 x 1,75 DIN 472	74186 - 1454	•	•	•	•	•	•	•	•	•	•	•	•	•	•	I , III
3.14	1	Ring	71457 - 1121	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3.15	2	Circlip 20 x 1,2 DIN 471	74186 - 1124	•	•	•	•	•	•	•	•	•	•	•	•	•	•	I , III , IV
3.16	1	Bush	71446 - 1151	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3.17	2	Grooved ball bearing 6004 DIN 625	71418 - 1254	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3.18	1	Retaining disk 30 x 42 x 1 DIN 988	74185 - 2094	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3.19	1	Sealing ring A 10 x 13,5 DIN 7603 - Cu	72712 - 1094	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3.20	1	Screw plug M 10 x 1 DIN 908	74106 - 1164	•	•	•	•	•	•	•	•	•	•	•	•	•	•	
3.21	1	Bush (not available as single part)	—															
3.22	1	Retaining disk 37 x 47 x 1 DIN 988	74185 - 2074					•	•	•	•							
3.23	2	Free-wheel boss	71161 - 1221					•	•	•	•							
3.24	10	Spring cup	71641 - 3111					•	•	•	•							I
3.25	10	Pressure spring	71613 - 2122					•	•	•	•							I
3.26	10	Roller	71961 - 2111					•	•	•	•							I
3.27	1	Circlip	74181 - 9111					•	•	•	•							
3.28	1	Washer	74181 - 5242					•	•	•	•							
3.29	1	Retaining disk S 20 x 28 x 2 DIN 988	74185 - 2214					•	•	•	•							
3.30	1	Ratched drive housing	71161 - 2241					•	•	•	•							
3.31	2	Hexagon-head nut M 8 DIN 934 - 8	74175 - 1284					•	•	•	•							
3.32	2	Lock washer 8 DIN 7980	74185 - 3044					•	•	•	•							
3.33	2	Cheese-head screw M 8 x 25 DIN 7984	74102 - 1254					•	•	•	•							
3.34	1	Lever	71532 - 1111					•	•	•	•							
3.35	1	Safety ring 22 x 1 DIN 472	74186 - 1274									•	•	•	•	•	•	II , III
3.36	1	Deep groove ball bearing 608 DIN 625	71418 - 1334									•	•	•	•	•	•	II
3.37	1	Circlip 8 x 0,8 DIN 471	74186 - 1024									•	•	•	•	•	•	II , III
3.38	1	Lid	71471 - 2212									•	•	•	•	•	•	II
3.39	3	Cheese-head screw M 6 x 16 DIN 7984	74102 - 1134									•	•	•	•	•	•	II
3.40	1	Gasket	72731 - 1812									•	•	•	•	•	•	II , III
3.41	1	Housing	74311 - 2631									•	•	•	•	•	•	II
3.42	1	Needle bearing 14 x 22 x 13	71418 - 1453									•	•	•	•	•	•	II
3.43	1	Radial sealing ring 14 x 22 x 4	72712 - 2173									•	•	•	•	•	•	II , III
3.44	1	Woodruff key 5 x 9 DIN 6888	71951 - 4054									•	•	•	•	•	•	I , II , III
3.45	1	Drive shaft (30 : 1)	71353 - 3121									•	•	•	•	•	•	II
		Drive shaft (60 : 1)	71353 - 3111														•	II
3.46	1	Screw plug M 10 x 1 DIN 908	74106 - 1164									•	•	•	•	•	•	II
3.47	1	Sealing ring A 10 x 13,5 DIN 7603 - Cu	72712 - 1094									•	•	•	•	•	•	II
3.48	1	Bush	71446 - 1161									•	•	•	•	•	•	IV
3.49	4	Cheese-head screw M 6 x 50 DIN 912	74102 - 0654									•	•	•	•	•	•	
3.50	4	Sealing ring A 6 x 10 DIN 7603	72712 - 1034									•	•	•	•	•	•	
3.51	1	Worm wheel (30 : 1)	71353 - 1821									•	•	•	•	•	•	II
		Worm wheel (60 : 1)	71353 - 1811														•	II
3.52	1	Gasket	72727 - 3232									•	•	•	•	•	•	II , III
3.53	1	Flange (not available as single part)	—															

LOOSE COMPONENTS GROUP (sub-assembly 4)

The loose components group contains all parts necessary for assembling of the sub-assemblies.

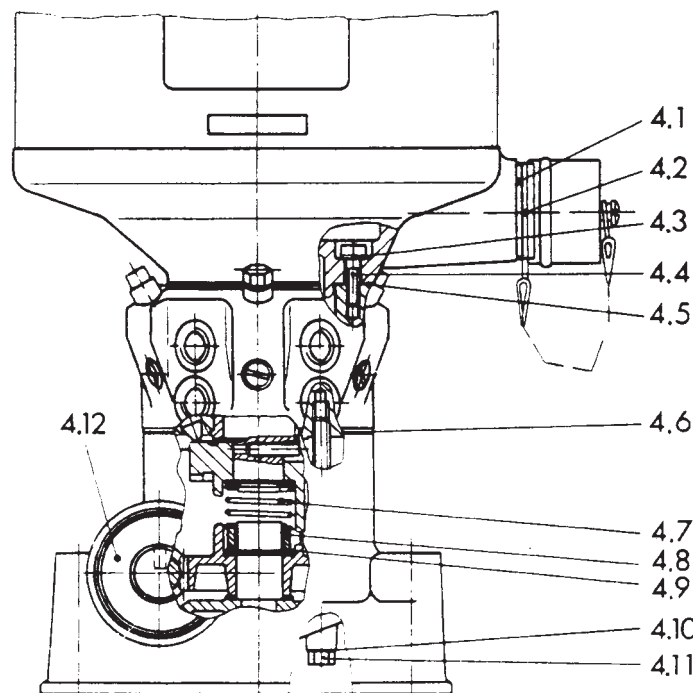


Fig. 9 loose components group for drive with free-end stud, ratched or reduction gear box

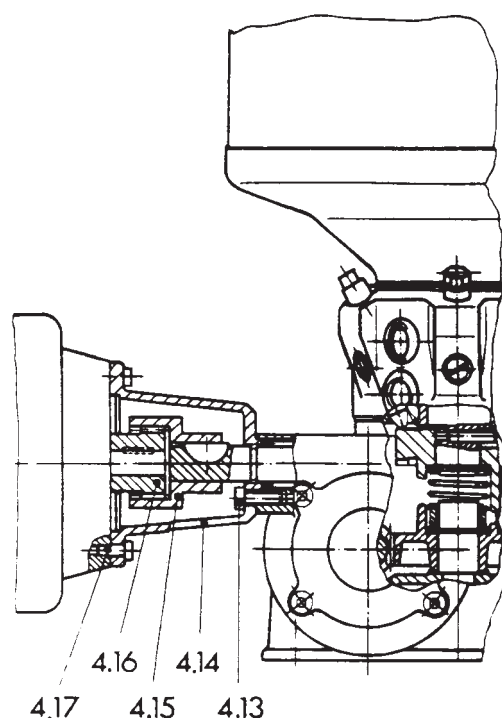


Fig. 10 loose components group for ratched drive with reduction gear box and integral flanged motor

Item	Pcs.	Denomination	Code no.	Re.
4. 0	1	Loose components groups for:		
		free-end stud	69111-7831	•
		Ratched drive or reduction gear drive	69111-7821	•
		Reduction gear with flanged motor	69111-7841	•
4. 1	1	Seal ring A 33 x 39 DIN 7603	72712-1624	• • • I,III
4. 2	1	Screw plug G 1 DIN908 phr	74106-1284	• • •
4. 3	3	Lock washer 6 DIN 7980	74185-3034	• • •
4. 4	3	Cheese-head screw M 6 x 20	74102-1144	• • •
4. 5	1	Gasket	72731-2412	• • • I,III
4. 6	1	Gasket	72727-3412	• • • I,III
4. 7	1	Pressure spring	71617-6151	• • •
4. 8	1	Cross piece	71941-2221	• • •
4. 9	1	Retaining disk 18 x 25 x 1	74185-2034	• • •
4.10	4	Seal ring A 6 x 10	72712-1034	• • • I,III
4.11	4	Cheese-head screw M 6 x 90	74102-1464	• • •
4.12	1	Cap	73761-3331	•
4.13	3	Cheese-head screw M 6 x 16	74102-1134	•
4.14	1	Flange	74354-5631	•
4.15	1	Sleeve	71931-1593	•
4.16	1	Hub	71931-1513	•
4.17	4	Cheese-head screw M 5 x 12	74102-1034	•

Remarks see page 7 - repair kits

THREE PHASE AC MOTOR (sub-assembly 5)

3phase AC flange-coupled motor
B 14 C 90 - 63 x 0.18 kW DIN 42677

Voltage	Frequency	Nominal RPM	Code no.
230 / 400 V	50 Hz	1500 min ⁻¹	76911 - 5133
265 / 460 V	60 Hz	1800 min ⁻¹	
500 V	50 Hz	1500 min ⁻¹	76911 - 5153

LOW LEVEL SWITCH (sub-assembly 6)

A level switch is available for the indication of the filling level in the 8 l; 15 l or 30 l tank. As sensing element serves an ultrasonic sensor. When the min. or max. allowed filling level is reached, a signal is given. With the help of a signal lamp which is mounted to the tank, this signal can be used for a visual warning or for the control of an automatic filling facility. In case of order, particular operating instructions are supplied for the level switch.

Reservoir	Code no.	
8 l	76951 - 6011	
15 l		
30 l		

FILLING VALVE (sub-assembly 7)

With automatic refilling over a bulk system the filler valve must be specifically ordered together with the pump.

Code no. _____ 65231 - 2111

SOLDERLESS SELF - TAPPING PIPE UNIONS (sub-assembly 8)

Denomination	Nominat size for pipe dia.	Code no.	Remarks
straight coupling	Ø 6	73442 - 1233	cylindrical stud
	Ø 8	73442 - 1324	
	Ø 10	73442 - 1414	
elbow coupling + straight coupling	Ø 6	73443 - 8123	The elbow union can only be used together with a straight screw-in fitting (take care that the nominal value is the same as that of the outside pipe diameter).
	Ø 6	73442 - 1233	
	Ø 8	73443 - 8133	
	Ø 8	73442 - 1324	
	Ø 10	73443 - 8193	
	Ø 10	73442 - 1414	
sealing ring A 14 x 18 DIN 7603-Cu		72712 - 1214	Every straight screw-in fitting requires a seal ring.

REPAIR KITS

I Pump with free-end stud and ratchet drive
Code no. _____ 69111 - 6851

II Transmission gear box

Ratio	Code no. drive position	
	1, 2, 4	5
30 : 1	61121 - 1511	61121 - 1531
60 : 1	61121 - 1521	61121 - 1541

III Pumps with reduction gear box and motor drive
Code no. _____ 69111 - 6861

IV Main transmission gear

Ratio 13 : 1 _____ Code no. 69111 - 6841
Ratio 17 : 1 _____ Code no. 69111 - 6881
Ratio 12 : 1 _____ Code no. 69111 - 6831
Ratio 25 : 1 _____ Code no. 69111 - 6821
Ratio 50 : 1 _____ Code no. 69111 - 6811

ATTENTION:

Repair kits see tables page 3, 5 and 6

The parts contained in repair kit are not available as single items.

SPECIFICATION

Permissible counter pressure : _____ 200 bar
and for a short period : _____ 250 bar

Permissible speed rotation of plunger

with rotary drive : _____ max. 10 RPM

with ratched drive : _____ max. 16 RPM

Max. number of lever strokes : _____ max. 300 min.-1

Consult us when rotation per minute exceed this rating or with RPM < 1, or when ZP-A or E 4 dividers are installed downflow of pump.

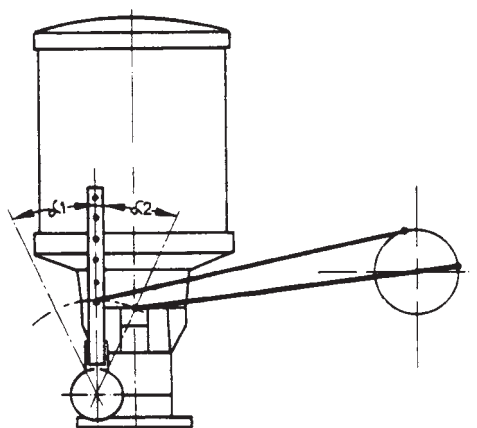


Fig. 11 angle with ratched drive

With ratchet drive the link work must be so arranged as to allow the deflection of the lever to be the same on both sides
 $\alpha 1 = \alpha 2 = \max. 50^\circ$

Max. oscillating lever swing : _____ 100 °

Min. oscillating lever swing : _____ 10 °

COMMISSIONING

Installation and connection of lubrication pump

Install and fix pump in vertical position. Connect motor with control box according to the wiring diagram. The direction of rotation of the motor may be either clockwise or anti-clockwise.

Recharge both lubricant reservoir and piping

Clean all piping carefully by tapping and purging and pre-charge with clean grease prior to installation using a grease gun.

When starting up for the first time, it is recommended to fill the grease reservoir up to the strainer with oil and then fill up with grease; otherwise the pump has to run for some time in order to bleed the system. Before connecting up the piping, run the pump at full speed until air-free lubricant emerges out of each outlet. Connect the lines to the pump using the seals supplied and run the pump until there is no inclusion of air-bubbles. Fill the bearings to be lubricated with grease and connect the lines to the lubrication points. The system is ready for commissioning as soon as the volume of lubricant to be delivered has been pre-set to the required rating.

Adjustment of feeds (fig 5)

At maximum delivery, corresponding to 0.1 ccm lubricant per outlet and plunger stroke, the mark on the outlet number is facing the fig. 4 stamped on the head of the adjusting spindle (2.5). Turning the spindle clockwise the output volume is reduced. Avoid reducing to below 1/4 of the max. volume (position 1). On pumps having 7 - 12 outlets, each adjusting spindle controls the output volume of two outlets, one on top of the other.

The arrow on the outlet number plate indicates the outlet ports associated with the respective adjusting spindle.

By changing the drive speed or the internal transmission ratio of the pump, or by changing the angle of the oscillating lever, the quantity delivered by each outlet, and consequently the total quantity can be varied.

MAINTENANCE

Recharging of lubricant container

As soon as about 3/4 of the contents of the reservoir have been used up, refilling should take place. The level of grease should never be allowed to drop so low as to expose the wedge plate, otherwise you may risk air finding its way into the feed lines. Efficient operation demands the use of only clean lubricant! Contamination of the lubricant can lead to operating trouble. The container must always be kept closed and care taken in ensuring that the lubricant is kept free from dirt and other contamination. Therefore containers should be recharged via the filler plug provided, preferably by use of a grease pump, or from the works bulk supply system.

Changing the number of outlets (fig. 5)

The pump will be supplied with the particular number of outlets specified on the order. If the circumstances of operation change, i.e. if the number of outlets has to be decreased or increased, proceed as follows:

To blind an outlet connect the pressure channel with the suction chamber. This is done by connecting the upper channels to the suction chamber by means of bore holes. To blind one of the top outlets remove the M 4 x 6 cheese-head screw (2.19). To re-commission a blinded outlet fit the cheese-head screw again, whereby taking care to use an A 4 x 8 ring (2.18) to ensure tightness. Generally close the appropriate outlets with the screw plug (2.17), or open it by removing the screw plug.

On a pump with paired outlets, one on top of the other, for instance, the lower outlet has to be put out of commission, too, the adjusting spindle (2.5) the ball (2.6) and the pressure spring (2.7) should be removed. Before taking out the spindle unscrew (2.9). Later insert the plug (2.20) into the bore of these spindles. To re-install the spindle after the trunnion screw (2.9) with sealing ring (2.8) has been screwed out, insert the ball followed by the spring. Both spring and ball are brought into their correct positions by the trunnion screw provided on condition that, when the screw is inserted, its head engages properly in the spiral groove of the adjusting spindle. It is recommended to grease the ball and the spring before insertion.

To blank a pair of outlets situated one above the other for a short period, set the spindle to 0 marking and make sure not to close the outlets firmly. Use stop plugs (2.20) instead of the locking screw plugs (2.17) to prevent the danger of overpressure when lubricant is delivered by accident.

ATTENTION :

An improper change of the number of outlets will result in damages to the pump.

Removal of drive shaft (fig. 6)

Remove in the following sequence :

Key (3.11) and cap (4.13) and circlip (3.13). After loosening the plug (3.20) the drive shaft (3.12) together with the ball bearing (3.17) and bush (3.16) with copper pin be dismantled by sticking against it with a sledgehammer.

Gear box :

The gear box is filled with the lubricant before leaving our works. Change lubricant only for dismantling and cleaning.

Filling screw :

main gear box	3.9
transmission gear box	3.46

recommended lubricant brands :

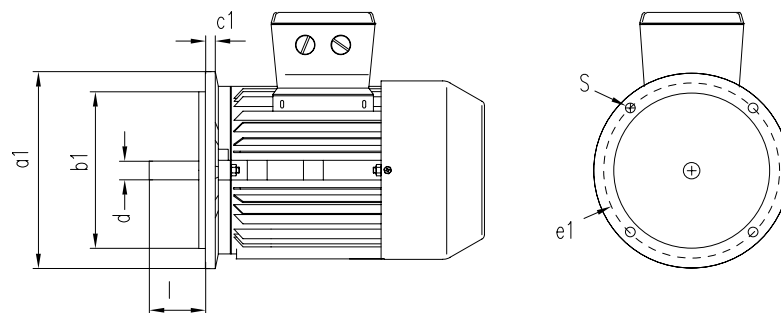
Aral Degol BG 680
BP Energol GR-XP 680
Calypsol fluid grease WIK 260 X (filled in our works)
Esso Spartan EP 680
Mobil Mobilgear 636
Shell Omala Oil 680
Texaco Meropa 680

MOTOR-INFORMATIONEN

MOTOR INFORMATION

INFORMATION DU MOTEUR

Delimon Bestellnr. Delimon Order-Nr. Delimon No de ref.	76911 - 5133		Anlauf Start Démarrage	direkt direct tout droit	
Hersteller Manufacturer Fabricant	Siemens		Einschaltung Switch on Mise en circuit	-	
Vorschriften Regulations Prescriptions	DIN VDE 0530 IEC 71		Isolierstoffklasse Insulating material class Class d'isolant	F	
Typ Type Type	1 LA7063-4AB12-Z		Sonderisolation Special insulation Isolation speciale	-	
Baugröße Size Taille	63		Bauform Model Modèles	IMB 14	
Frequenz Frequency Fréquence	50Hz	60Hz	Gewicht Weight Poids	4,1 Kg	
Nennleistung Power rating Puissance nominal	N	0,18kW	0,21kW	Gehäuseschutzart Enclosure protection Degré de protection	IP 55
Nenndrehzahl Nominal speed Vitesse nominal	n	1320/min	1620/min	Explosionsschutz Flame-proof enclosure Protect. anti-déflagrante	-
Nennspannung Voltage Tension nominal	U	230V/400V ±10%	460V ±10%		
Ständerschaltung Stator circuit Couplage du stator		Δ/Y	Y		
Nennstrom Nominal current Intensité nominal		1,01/0,59A	0,58A	Klemmkasten Conduit box Boît à bornes	
Spannungs- und Strombereich Voltage and current range Tension et Intensité		220-240V ±5% 1,03-1,01A 380-420V ±5% 0,6-0,58A	440-480V ±5% 0,6-0,58A	Lage Position Position	oben above en haut
Leistungsfaktor Power factor Facteur de puissance	cosφ	0,77	0,76	Schutzart Protection Protection	IP 55
Wirkungsgrad Efficiency Rendement	η	60%		Verschraubung Cable entry Entrée de cable	1 x M 25 x 1,5 1 x M 16 x 1,5
Rel. Anlaufstrom Starting current ratio Courant rel. de démarrage	I _a /I _n	3,2		Kabelquerschnitt Cable cross section Section de cable	2,5 mm²
Rel. Anlaufmoment Starting torque ratio Couple rel. de démarrage	M _a /M _n	2,0		Klemmenanzahl Number of terminals Nombre de bornes	6
Rel. Kippmoment Pull-out torque ratio Couple rel. de décrochage	M _k /M _n	2,0			
Nenndrehmoment Nominal torque Décrochage nominal	M _n	1,3 Nm		Flanch Flange Bride	a1:90, b1:60, c1:2,5, d:11, e1:75, l:23, s:M5



Level Switch 76951 – 6011

Description for Programmer

Field of application:

The level control device is used for the control of up to three filling level in tanks.

Features : Switching interval adjustable from 60 mm to 550 mm

Operating voltage range 20 V to 252 V AC / DC.

Relay outputs.

Parameterizable via DIP switch.

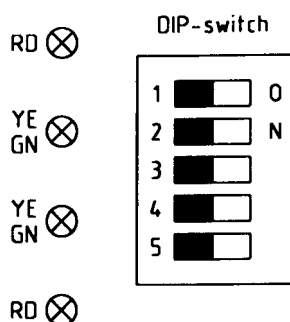
Installation position:

The level switch is installed vertically to the surface of the liquid in the lid of the tank.

Teaching-in of the switching points:

How to prepare the teaching-in of the switching points:

- Remove transparent lid and insertion plate
- Switch on operating voltage



- Switch DIP switch 5 in ON position. The red LED's 1 and 4 will light up.
During the teaching-in process, the relays switch into the dead state.
- Position suitable target in the desired switching interval

Teaching-in of switching point 1:

- Switch over DIP switch 1.
If the target is recognized, a green LED will light up additionally.
- If the target is not recognized, a yellow LED will light up additionally.
The target has to be positioned a new until it is recognized.
- Switch back DIP switch 1.
The switching interval is stored in the RAM memory when the DIP switch is switched back while the green LED is flashing. After this, only the red LED's will emit light.
This state shows the operator that the DIP switch is in its initial position again.

Teaching-in of switching point 2:

- Please refer switching point 1, but switch over DIP switch 2.

Teaching-in of switching point 3:

- Please refer switching point 1, but switch over DIP switch 3.

How to finish the teaching-in process :

- Switch DIP switch 5 back into OFF position.
The adjustment will be transferred into the (retentive) memory with permanent effect.

After the teaching-in, the order of the switching points is always
switching point 1 < switching point 2 < switching point 3.

- Insert insertion plate according to the position of DIP switch 4.
- Screw on transparent lid and insertion plate.

Teaching-in of the default values :

- Switch DIP switch 5 in ON position. The red LED's 1 and 4 will light up.
- Switch over DIP switch 4.
- Switch over DIP switch 1, then switch it back again.
- Switch over DIP switch 2, then switch it back again.
- Switch over DIP switch 3, then switch it back again.
- Switch DIP switch 4 back again.
- Switch DIP switch 5 back into OFF position again.
The default adjustments are transferred into the memory with permanent effect.

The following default values have been adjusted : S1 = 60mm; S2 = 220mm; S3 = 270mm
(measured from the sensor to the switching point)

Adjustment of the switching behaviour :

In normal operation (DIP switch 5 in OFF position), one can adjust the switching behaviour of switching outputs S1 to S3 by means of the DIP switches 1 to 3.

DIP switch	ON position	OFF position	<i>State ipon delivery</i>
DIP switch 1	NO contact behaviour	NC contact behaviour	OFF position
DIP switch 2	NO contact behaviour	NC contact behaviour	OFF position
DIP switch 3	NO contact behaviour	NC contact behaviour	OFF position

“NO (normally open) contact behaviour“ means : The corresponding contact closes, when the filling level is higher than the pertaining switching point.

“NC (normally closed) contact behaviour“ means : The corresponding contact opens, when the filling level is lower than the pertaining switching point.

Adjustment of the indication mode :

In normal operation, (DIP switch 5 in OFF position), one can choose the indication mode by means of DIP switch 4.

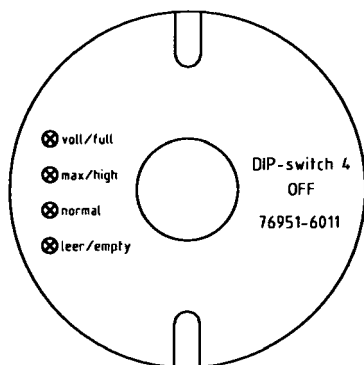
Mode “early warning/max. filling level” (state upon delivery)

DIP switch 4 in OFF position

- LED 1 (red): overfill signal (full) switching point S1
- LED 2 (yellow) early warning (max) switching point S2
- LED 3 (green) normal state (normal) -
- LED 4 (red) empty signal (empty) switching point S3

Filling level x	$x > S1$	$S1 > x > S2$	$S2 > x > S3$	$x < S3$
LED 1, red	flashes	off	off	off
LED 2, green/yellow	off	flashes yellow	off	off
LED 3, green/yellow	off	off	flashes green	off
LED 4, red	off	off	off	flashes

Insertion plate :



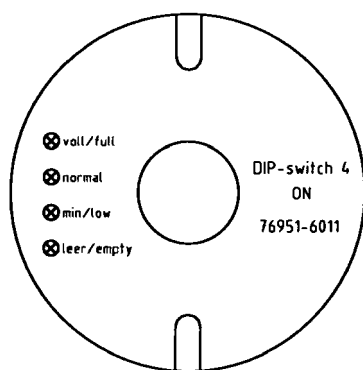
Mode “early warning/min. filling level“

DIP switch 4 in ON position :

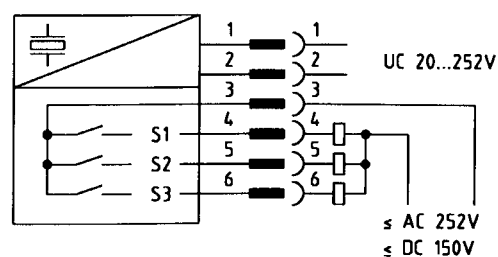
- LED 1 (red): overfill signal (full) switching point S1
- LED 2 (green) normal state (normal) -
- LED 3 (yellow) early warning (min) switching point S2
- LED 4 (red) empty signal (empty) switching point S3

Filling level x	$x > S1$	$S1 > x > S2$	$S2 > x > S3$	$x < S3$
LED 1, red	flashes	off	off	off
LED 2, green/yellow	off	flashes green	off	off
LED 3, green/yellow	off	off	flashes yellow	off
LED 4, red	off	off	off	flashes

Insertion plate:

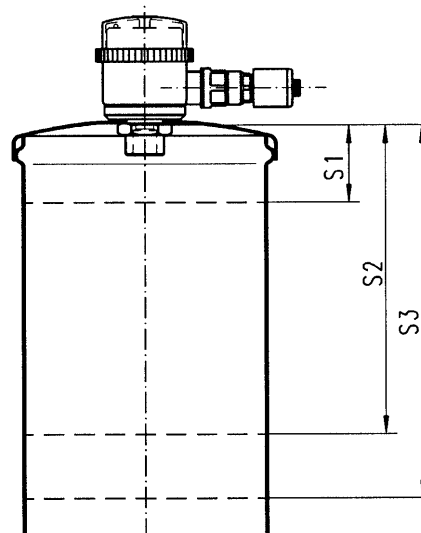


Connection diagram:



Adjustment:

Pump	Reservoir	S 1 mm	S 2 mm	S 3 mm
FZ	8 l	60	240	290
FZ	15 l	60	265	315
FZ	30 l	60	335	385
BM – B	8 l	60	240	290
BM – B	15 l	60	265	315
BM – B	30 l	60	335	385
BS – B	30 l	60	335	385
BS – B	60 l	60	390	440
BS – B	100 l	60	480	530
Autolub	8 l	60	220	270
HA III	100 l	60	460	510



Specification:

General data:

Acquisition range	60 mm...550 mm (ex ultrasonic transducer)
Reference quantity	100 mm x 100 mm (even surface)
Opening angle of sound beam	approx. 5° at -6dB sound pressure
Transducer frequency	approx. 380 kHz
Repeat accuracy	< 0,5 %, < 0,5 mm at constant ambient conditions
Reaction time of relay contacts	approx. 10 s
Release time of relay contacts	approx. 10 s
Reaction time of LEDs	< 1 s (indication of measured value)
Release time of LEDs	< 1 s (indication of measured value)
Temperature drift	< 4 % over the whole temperature range

Electrical data:

Operating voltage U_B	20 V...252 V DC/AC, 47 Hz...63 Hz
Current consumption	ca. 70 mA at 30 V DC (all relays in working position)
Switching output 1, 2 and 3	relays
• Contact material	gilded silver alloy
• Switching voltage max.	150 V DC, 252 V AC
• Switching current max.	3 A, ohmic
• Switching current min.	0,1 mA, 100 mV DC
• Switching capacity	90 W at 30 V DC, 750 W at 250 V AC
• Life time (electrical)	10 ⁶ switching cycles at a resistive load of 3 A / 30 V DC and/or 3 A / 252 V AC

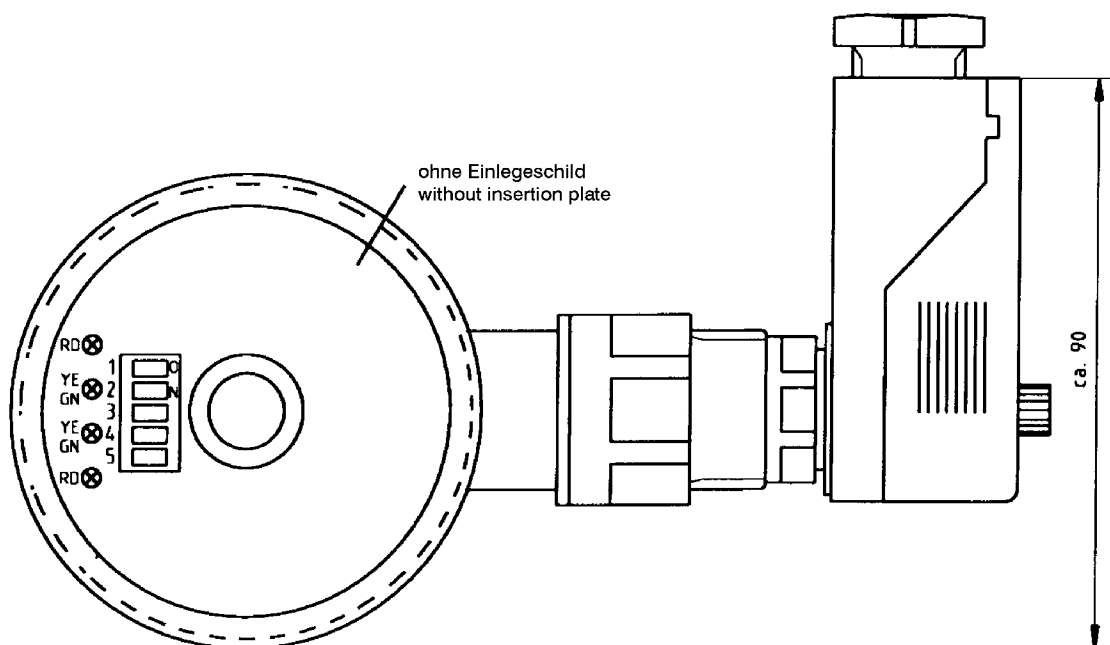
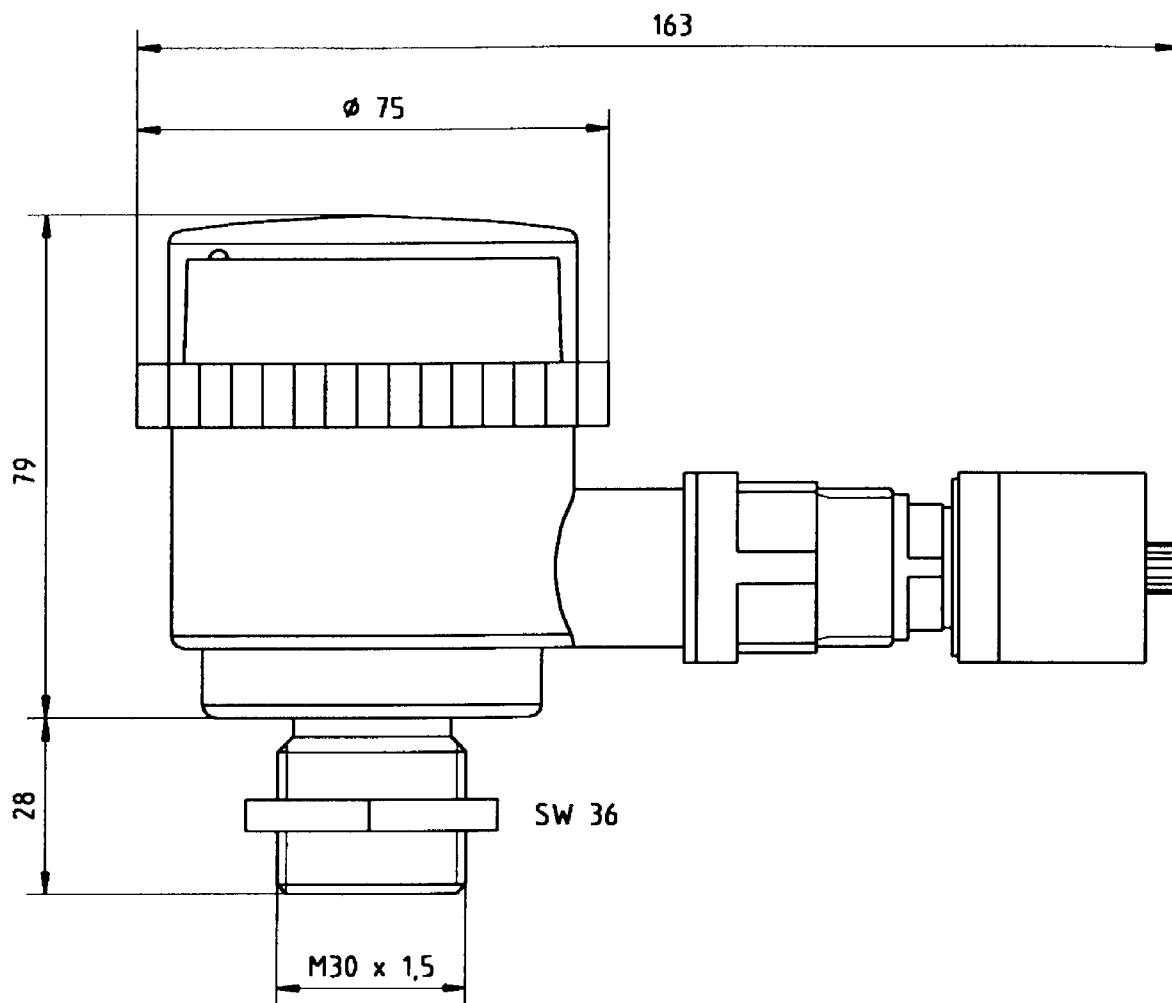
Mechanical data:

Operating temperature	-20 °C...+60 °C
Storage temperature	-40 °C...+85 °C
Protective system acc. DIN 40 050	IP 65
Type of connection	7-polar 90°- cable box (PG 13,5) with screw and soldering contacts Cross section: max. 1,5 mm ² wire sleeves without insulation collar.
Weight	0.55 kg

State upon delivery

DIP switch	switch 1-5 in OFF position.
Indication mode	indication with early warning "filling level max."

Dimensioned drawing:



Replacement plug for level switch 76951-6011: Code no. 76951-6021

Druckbegrenzungsventil NU - A

Pressure relief valve NU - A

ANWENDUNG

Das Druckbegrenzungsventil NU-A wird vorzugsweise als Sicherheitsventil in Mehrleitungs- und Progressiv-Anlagen eingesetzt. Die Zentralschmieranlagen werden durch das Druckbegrenzungsventil NU-A vor Überdruck gesichert.

APPLICATION

The NU-A type pressure relief valve is used as safety valve to protect central lubrication systems from overpressure. It is suitable especially in multi-line systems and progressive systems.

MERKMALE

- Es kann für alle Arten von Zentralschmieranlagen verwendet werden, soweit diese im Druckbereich von 15 bis 300 bar arbeiten.
- Einfacher Aufbau, dadurch preisgünstig, wartungsfrei, leicht einstellbar.
- Geringe Abmessungen

CHARACTERISTICS

- *It can be used for all kinds of centralized lubrication systems as far as the latter work in the pressure range between 15 and 300 bar.*
- *Simple configuration, thus low price, maintenance-free, and easy to set.*
- *Reduces space.*

TECHNISCHE DATEN

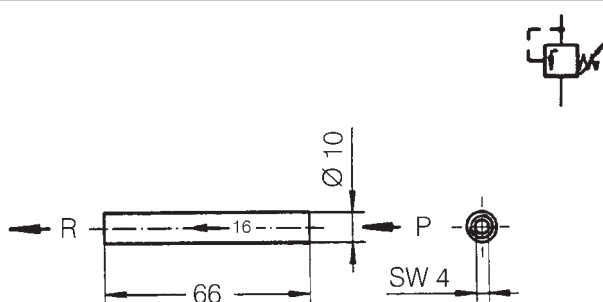
Öffnungsdruck : _____ einstellbar 15 bis 300 bar
 Betriebstemperatur : _____ - 20 °C bis + 120 °C
 Durchflußmedium:
 Schmierfette _____ auf Mineralölbasis NLGI-Klasse 000 bis 3 DIN 51818
 Mineralöle _____ mit einer Viskosität über 68 mm²/s bei Betriebstemperatur
 Einbaulage : _____ beliebig
 Gewicht : _____ 0,03 kg

TECHNICAL DATA

Response pressure : _____ adjustable from 15 up to 300 bar
 Operating temperature : _____ - 20 °C to + 120 °C
 Flow media :
 Lubricant greases _____ based on mineral oil NLGI-class 000 to 3 DIN 51818
 Mineral oils _____ with a viscosity above 68 mm²/s at operating temperature
 Installation : _____ at choice
 Weight : _____ 0.03 kg

Einstellbereich Setting range	eingestellt auf set to	Durchfluß- medium Flow media	Bestell - Nr. Code no.
15 - 70 bar	16 bar	Fett oder Öl grease or oil	38132 - 1211
15 - 70 bar	70 bar	Fett oder Öl grease or oil	38132 - 1221
50 - 200 bar	160 bar	Fett oder Öl grease or oil	38132 - 1231
100 - 300 bar	200 bar	Fett grease	38132 - 1241
100 - 300 bar	300 bar	Fett grease	38132 - 1251

ABMESSUNGEN / DIMENSIONS (mm)

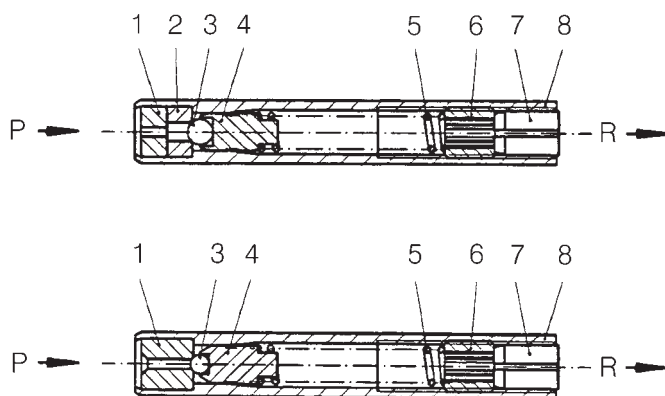


WIRKUNGSWEISE

Tritt in der Zentralschmieranlage ein Druck auf, der über dem eingestellten Wert liegt, wird ein federbelasteter Kolben (4) angehoben, wodurch der Schmierstoff ins Freie oder über eine Rücklaufleitung zum Pumpenbehälter bzw. zum Saugraum der Pumpe zurückfließt.

MODE OF OPERATION

In the event of a pressure in a central lubrication system rising the excess of the pre-set rating, the spring-loaded piston (4) is raised thus allowing the lubricant to escape into the open air or to flow back via a return line into the pump reservoir or into the pump suction chamber.

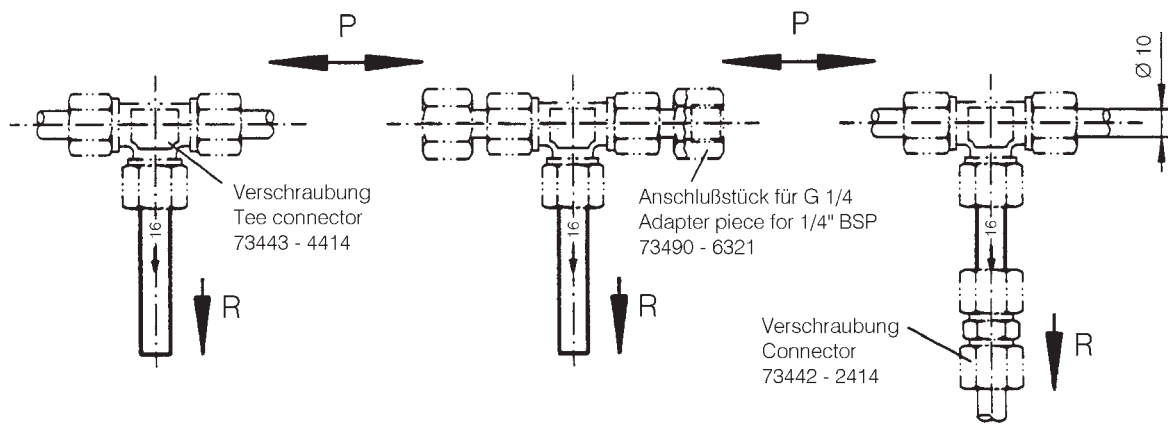


Darstellung entspricht
Bestell - Nr.
Picture corresponds to
code no.
38132 - 1211
38132 - 1221
38152 - 1231

Darstellung entspricht
Bestell - Nr.
Picture corresponds to
code no.
38132 - 1241
38132 - 1251

EINBAUVORSCHLÄGE

SUGGESTIONS FOR INSTALLATION



EINBAU

Zum Einbau wird in die vor Überdruck zu schützende Druckleitung eine T-Verschraubung für Rohr Ø 10 eingesetzt und in den freien Anschluß das Druckbegrenzungsventil so eingesetzt, daß die Einstellschraube sichtbar bleibt.
Wird eine Rücklaufleitung vorgesehen, ist diese erst zu befestigen, wenn der Ansprechdruck eingestellt ist.

INSTALLATION

*Into the pressure line to be protected against overpressure a Tee connector for tube 10 mm dia. will be fitted; the relief valve is to be located towards the free connection as to ensure that the setting screw remains visible.
In lubrication systems where a return line is provided, connect the line only when the response pressure rating has been set.*

SICHERHEITSHINWEISE

ACHTUNG

Den Öffnungsdruck des Druckbegrenzungsventils nie höher einstellen als den maximal zulässigen Betriebsdruck der Zentralschmieranlage.



Der Schmierstoff kann je nach Anlagendruck - wenn das Druckbegrenzungsventil öffnet - unter hohem Druck austreten. Deshalb muß, wenn an Anschluß "R" keine Rücklaufleitung angeschlossen ist, das Druckbegrenzungsventil so justiert werden, daß seine Öffnung nicht auf Personen, Arbeitsplätze und Maschinen gerichtet ist.

Ist durch Justierung des Druckbegrenzungsventils keine ausreichende Sicherheit für Personen, Arbeitsplätze und Maschinen zu erreichen, muß am Anschluß "R" des Druckbegrenzungsventils ein Ableitrohr montiert werden.

Bei Nichtbeachtung der Arbeitssicherheits-Hinweise besteht Gefahr für Leib und Leben von Personen.

SAFETY ADVICE

ATTENTION

Never adjust the opening pressure of the pressure relief valve to a higher value than the max. allowed service pressure of the centralized lubrication system.



Depending on the system pressure, the lubricant may - when the pressure relief valve opens - escape under high pressure. Therefore it is essential, if there is no return line connected to connection "R", to adjust the pressure relief valve in such a way that its opening is not directed towards persons, working places or machines.

If it is impossible to obtain a sufficient security for persons, working places and machines by adjustment of the pressure relief valve, a drain pipe has to be mounted at connection "R" of the pressure relief valve.

The nonobservance of the working safety advice means danger to life and limb of persons.

EINSTELLUNG

Zur Erhöhung des Einstelldruckes ist die Innensechskantschraube (6) (SW 4) im Uhrzeigersinn zu drehen, wobei die Feder (5) in der Hülse (8) gespannt wird. Nach Beenden der Einstellung muß der Spannstift (7) mit einem Dorn soweit in die Hülse (8) geschoben werden, bis er auf der Innensechskantschraube (6) aufliegt.

Zur Verringerung des Einstelldruckes ist die Innensechskantschraube (6) (SW 4) entgegen dem Uhrzeigersinn zu drehen, wobei die Feder (5) in der Hülse (8) entspannt wird.

SETTING

To increase the set pressure, turn the hexagon socket screw (6) (wrench size across flats 4) clockwise. This tensions the spring (5) in the sleeve (8). When the adjustment is finished, push the tension pin (7) with a drift into the sleeve (8) until it lies on the hexagon socket screw (6).

To decrease the set pressure, turn the hexagon socket screw (6) (wrench size across flats 4) anticlockwise. This release the spring (5) in the sleeve (8).

ZUBEHÖR

Verschraubungen, Anschlußstücke und Rohre müssen gesondert bestellt werden.

ACCESSORIES

Union pieces, connection pieces and pipes have to be specified separately on the order.

Progressive Distributor ZP – A

USE

The major element of a progressive system is the lubricant distributor.

ZP - A distributors are used to divide and meter the lubricant in total loss central lubrication systems (oil, semi-fluids and greases) and in oil recirculation systems suitable for small, medium and large machine plants.

ADVANTAGES

- ☐ easy system layout
- ☐ problem-free installation
- ☐ unsophisticated control and monitoring features
- ☐ precisely-measured lubricant discharge per outlet
- ☐ variable flow rates per element suit individual requirements

APPLICATIONS

Machine - tools and processing machinery, mechanical engineering in general, presses of every type, plastic - and paper processing machines, textile machines, printing - and packaging machinery, etc.

DESIGN

The distributors consist of several individual elements (minimum 3 in a standard array) which are screwed together and sealed to each other.

Depending upon the arrangement within the distributor the following segment types are available.

Initial -	or	A - element
Medium	or	M - element
End -	or	E - element



CONFIGURATION AND FUNCTION

The major element of a progressive system is the lubricant distributor, whose function is to divide a lubricant volume delivered under pressure (oil and grease) successively into smaller part volumes towards the 24 or more (upon request) of possible outlets. This is achieved thanks to the pistons incorporated in progressive distributors which are positively displaced and controlled against each other.

Pressurized lubricant entering forces the feed pistons to be moved in a certain sequence to the end of the strokes into their initial positions, thus delivering a full measures charge of lubricant to the outlets, one after another. The distributor remains operating as long as lubricant is being delivered.

Upon request, ZP-A distributors are supplied with integrated non-return valves located in the outlet bore; this embodies the advantage to ensure a reliable operation however small a lubricant is being discharged, and at higher counterpressure situations to overcome with flexible pipework installed downflow.

In addition, we supply distributor types with motion indicators where an electric limit switch and/or electronic monitor may be fitted enabling external warning and monitoring function.

TECHNICAL DATA

Max. operating pressure : _____ 160 bar
 Temperature range : _____ – 20 °C up to + 80 °C (if high temperatures are prevailing consult us)
 Admissible differential pressure between 2 outlets : _____ max. 50 bar
 _____ with non-return valves max. up to admissible system pressure
 Output volume each piston stroke : _____ 0.07; 0.1; 0.2; or 0.3 ccm
 Flow volume for oil and grease : _____ min. 0.5 ccm/min; max. 1000 ccm/min
 Non-return valve opening pressure : _____ 2 bar
 Response pressure : _____ ≤ 10 bar
 Compatible lubricants on mineral oil basis:
 Grease lubricants up to _____ NLGI class 3 DIN 51818
 Oil _____ ISO VG 68 to 1500 (DIN 51519) at 20 °C ambient temperature
 Synthetic lubricants _____ at request
 Pipe connection:
 Inlet port _____ Ø 6 or Ø 8
 Outlet port _____ Ø 6

max. bar outlets	No. of elements	Dimensions		Weight approx. kg	Standard	c/w non-return valve	c/w 2 motion indicators	c/w non-return valve with 2 motion indicators
		a	b		Code no.	Code no.	Code no.	Code no.
6	3	48	—	1.03	35541 - 1031	35541 - 2031	35541 - 3031	35541 - 4031
8	4	64	—	1.37	35541 - 1041	35541 - 2041	35541 - 3041	35541 - 4041
10	5	80	—	1.72	35541 - 1051	35541 - 2051	35541 - 3051	35541 - 4051
12	6	96	137	2.06	35541 - 1061	35541 - 2061	35541 - 3061	35541 - 4061
14	7	112	153	2.40	35541 - 1071	35541 - 2071	35541 - 3071	35541 - 4071
16	8	128	169	2.75	35541 - 1081	35541 - 2081	35541 - 3081	35541 - 4081
18	9	144	185	3.09	35541 - 1091	35541 - 2091	35541 - 3091	35541 - 4091
20	10	160	201	3.43	35541 - 1101	35541 - 2101	35541 - 3101	35541 - 4101
22	11	176	217	3.77	35541 - 1111	35541 - 2111	35541 - 3111	35541 - 4111
24	12	192	233	4.11	35541 - 1121	35541 - 2121	35541 - 3121	35541 - 4121

Attention:

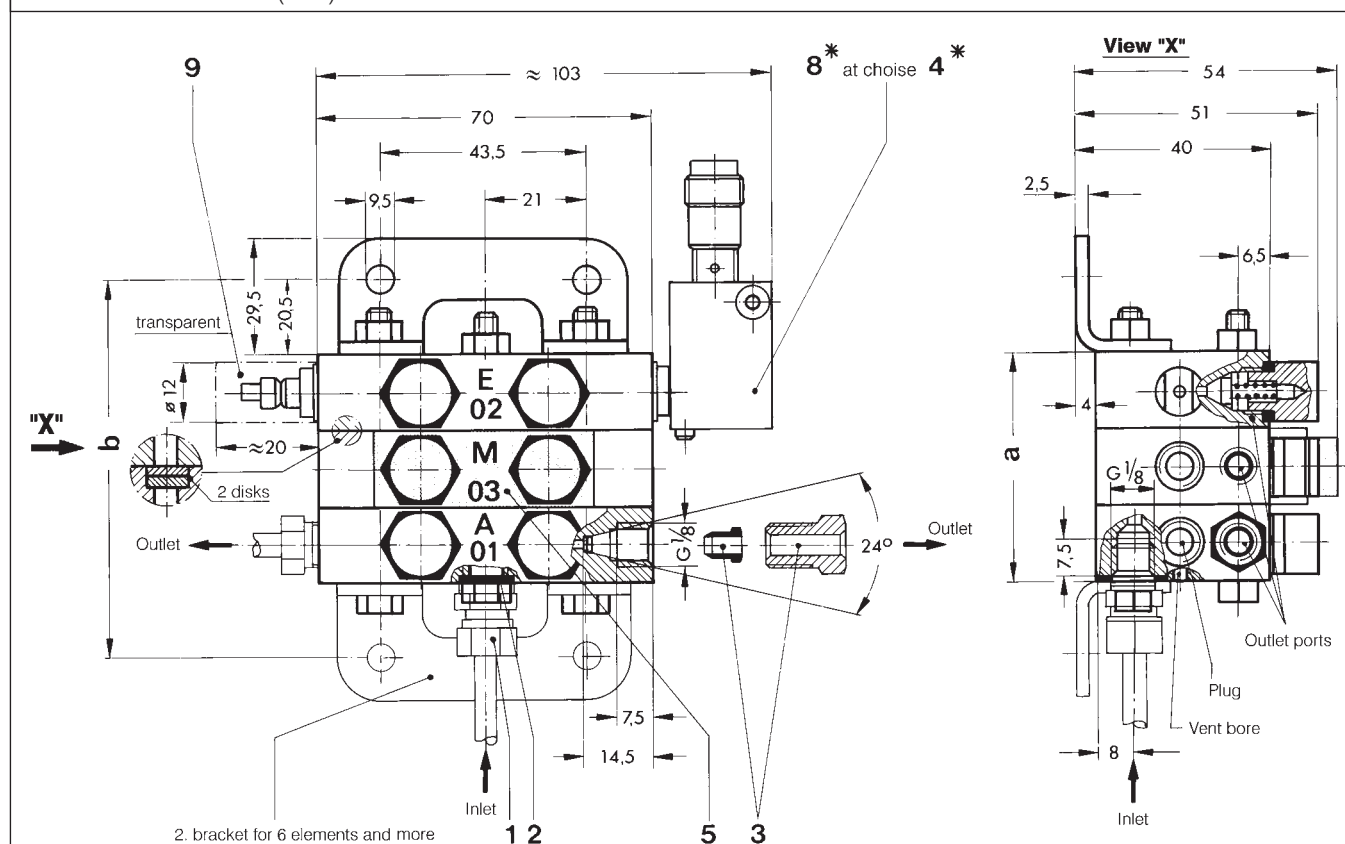
It must be remembered that the lubricant metered by a given piston does not exit out of same element but in the adjacent element next to the inlet port. The quantity metered by the piston provided in the initial element is discharged out of the end element outlet port.

ACCESSORIES (to be ordered separately)

Part	Denomination	Code no.
1	Straight male stud connector for pipe O.D. $\varnothing 6 *$	73442 – 1214
	$\varnothing 8 **$	73442 – 1314
2	Copper washer A 10 x 13.5 DIN 7603-Cu	72712 – 1094
3	Connector for pipe O.D. $\varnothing 6$ brass	63422 – 2241
	steel	63422 – 2261
4	Limit switch	66922 – 2211
	max. voltage 500 V AC 24 V DC 220 V DC	
	max. current 10 A — —	
	power — 10 W 4 W	
	enclosure IP 65 max. frequency 1 Hz	
5	Bridge	63421 – 2211
8	Proximity switch (electronic PNP)	66925 – 1311
	Voltage range 10 - 30 V DC	
	max. current 200mA	
	max. frequency 1000 Hz	
	enclosure IP 65	
	Plug with cable see data sheet 66925 - 1311 M1	
9	Protection cap for indicator pin (silicon)	73649 – 1811
—	over-pressure indicator * see data sheet 65112 - 1300 M1	
	* only available with plug	73490 – 3021
	* only in conjunction with bridge part 5 hollow screw	74111 – 3811

* thread: cylindrical ** thread: conical

DIMENSIONS (mm) Distributor with non-return valves and motion indicators



* Part 4 = limit switch and part 8 proximity switch are located on the right hand side normally. Both switch types, however, can also be attached, at choice, on the left hand side.

EXPLANATION

- The basic configuration of distributor ZP-A is illustrated by a pictorial symbol; the channel bores marked are to indicate that the metering volume in one element principally is discharged out of the adjacent element located next to the 'inlet' port, with one exception; the metering volume in the initial element exits only out of the end element.

007 is equal to 0.07 ccm/per piston stroke
 01 is equal to 0.1 ccm/per piston stroke
 02 is equal to 0.2 ccm/per piston stroke
 03 is equal to 0.3 ccm/per piston stroke

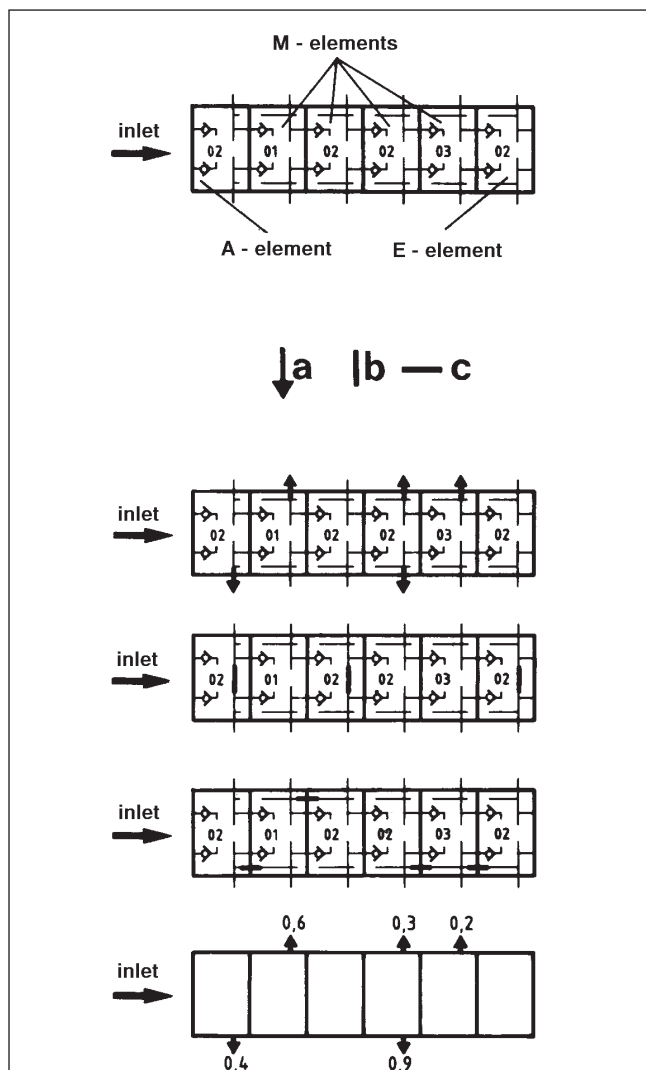
- There are various means to combine several metering volumes to be delivered to a single outlet port, the following 3 letter symbols are used to mark it accordingly.

Symbol "a" to denote the position of the outlet port.

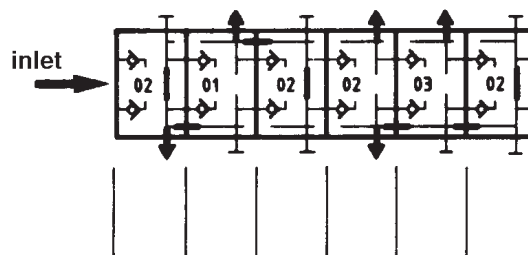
Symbol "b" to denote longitudinal cross-porting of both volumes within one element, by fitting a bridge on the respective element.

Symbol "c" to denote transverse cross-porting by combining the metering volumes in neighbouring elements which necessitates that the seals between elements in direction to the inlet port are removed and the unused outlet blanked by plug (this combination is not feasible in the initial element).

- Metering volume at outlet port (ccm).



EXAMPLE FOR ORDERING



Denomination:

35541 - 2061 / 02 ba 01 ac 02 cb 02 aa 03 ac 02 bc

Code No. classification:

Type ZP-A

with non-return valve

number of elements

metering volume (element size)

discharge of upper metering volume

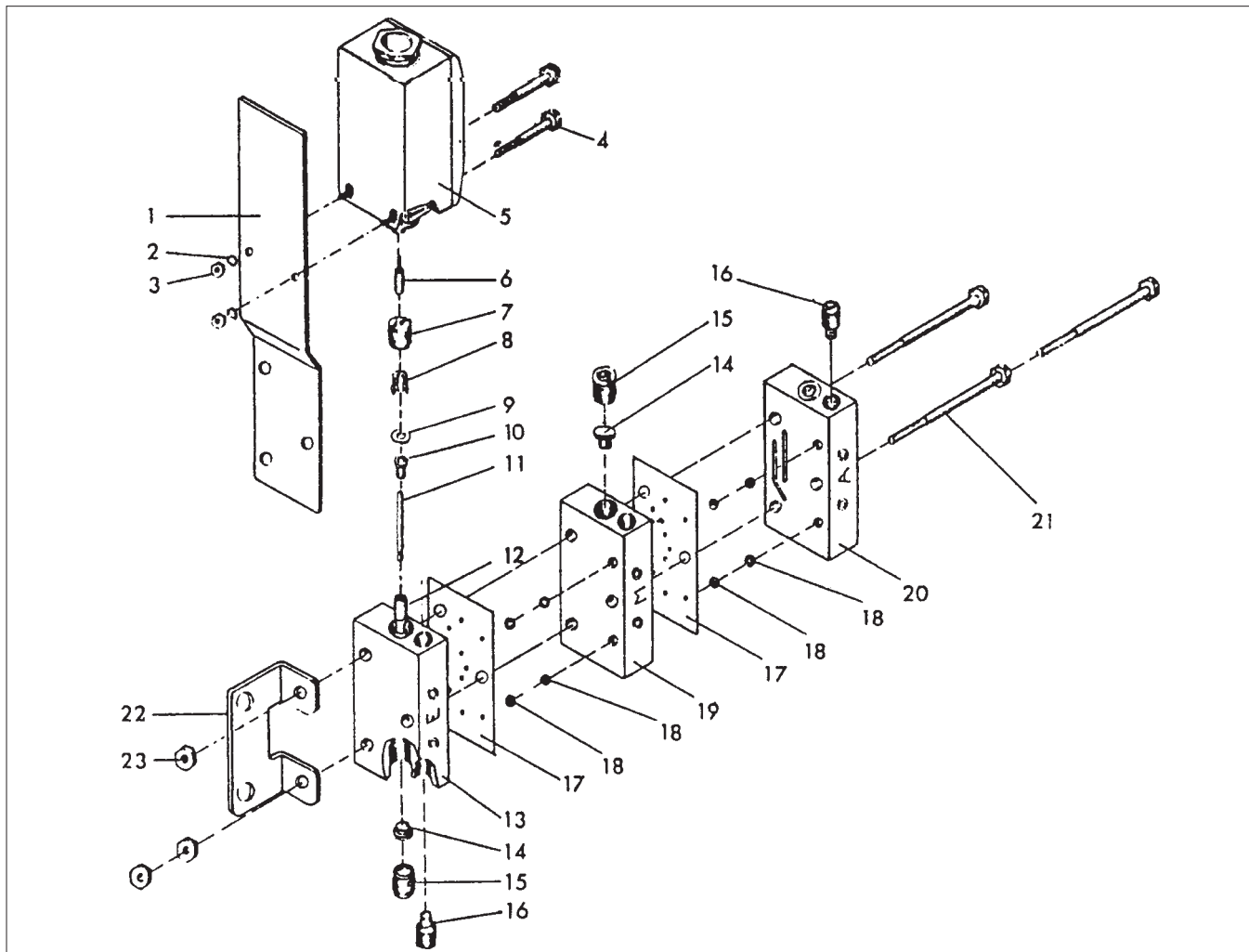
discharge of lower metering volume

Schmierstoff - Verteiler ZP - A

Ersatzteilzeichnung und Ersatzteilliste

Lubricant distributor ZP - A

Spare parts drawing and Spare parts list



Pos. Item no.	Benennung Denomination	Bestell - Nr. Code no.								
--	Enttaster / Limit switch	66922 - 2211	•							
--	E-Segment / E-segment	64334 - 2631		•						
--	M-Segment / M-segment	01 64334 - 2671			•					
	02 64334 - 2621					•				
	03 64334 - 2651						•			
--	A-Segment / A-segment	64334 - 2611						•		
1	Winkel / Angle	74361 - 1171	•							
2	Federring 4 DIN 7980 / Spring washer 4 DIN 7980	74185 - 3014	•							
3	Sechskantschraube M 4 DIN 934 / Hexagon nut M 4 DIN 934	74175 - 1234	•							
4	Zylinderschraube M 4 x 35 DIN 912 Cheese-head screw M 4 x 35 DIN 912	74102 - 1674	•							

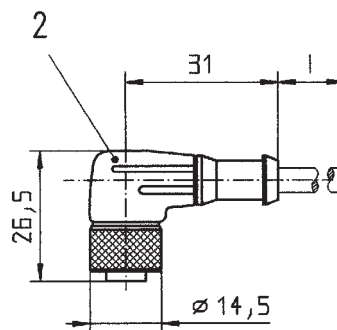
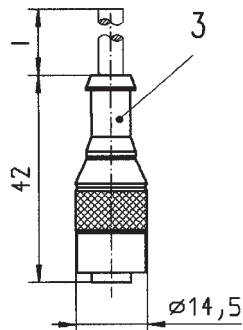
Pos. Item no.	Benennung Denomination	Bestell - Nr. Code no.							
5	Endtaster / <i>Limit switch</i>	76925 - 5123	•						
6	Gewindestift / <i>Headless pin</i>	74141 - 1921	•						
7	Verschraubung / <i>Screw joint</i>	74161 - 6621		•					
8	Nutring 7,45 / <i>Groove ring 7,45</i>	72713 - 9093		•					E
9	Scheibe / <i>Washer</i>	74181 - 1621		•					
10	Überwurfschraube / <i>Cap nut</i>	73400 - 3121		•					
11	Bewegungszeiger / <i>Motion indicator</i>	64161 - 6111		•					
12	Kolben / <i>Piston</i>	Ø 5			•				N
		Ø 7		•		•		•	N
		Ø 8						•	N
13	E-Segmentgehäuse / <i>E-segment housing</i>	74331 - 9831		•					N
14	Stopfen / <i>Stopper</i>	71451 - 9111		•	•	•	•	•	
15	Verschlußschraube / <i>Plug screw</i>	74161 - 7911		•	•	•	•	•	
16	Verschlußschraube / <i>Plug screw</i>	74161 - 5421							
17	Dichtung / <i>Sealing point</i>	72731 - 3512							E
18	Verschlußscheibe 5 DIN 470 / <i>Plug washer 5 DIN 470</i>	74185 - 2324							E
19	M-Segmentgehäuse Kennzeichen <i>M-segment housing marking</i>	01			•				N
		02				•			N
		03						•	N
20	M-Segmentgehäuse / <i>M-segment housing</i>	74331 - 9811						•	N
21	Sechskantschraube DIN 931 <i>Hexagon head nut DIN 931</i>	M 6 x 60 (für 3 Segmente) <i>(for 3 segments)</i>							
		M 6 x 75 (für 4 Segmente) <i>(for 4 segments)</i>							
		M 6 x 90 (für 5 Segmente) <i>(for 5 segments)</i>							
	Siftschraube / <i>Stud screw</i>	(für 6 Segmente) <i>(for 6 segments)</i>							
		(für 7 Segmente) <i>(for 7 segments)</i>							
		(für 8 Segmente) <i>(for 8 segments)</i>							
		(für 9 Segmente) <i>(for 9 segments)</i>							
		(für 10 Segmente) <i>(for 10 segments)</i>							
		(für 11 Segmente) <i>(for 11 segments)</i>							
		(für 12 Segmente) <i>(for 12 segments)</i>							
22	Winkel / <i>Angle</i>	74361 - 4241							
23	Sechskantmutter M 6 DIN 934 / <i>Hexagon nut M 6 DIN 934</i>	74175 - 1254							

Die mit "E" gekennzeichneten Teile werden zur Ersatzteilhaltung empfohlen.
Die mit "N" gekennzeichneten Teile werden nicht einzeln als Ersatzteile geliefert.

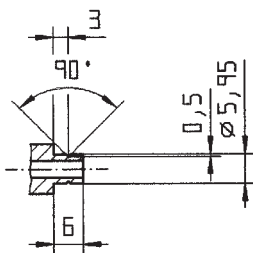
*The parts marked "E" are recommended to be carried in stock.
The parts marked "N" are not available by retail.*

Monitoring switch

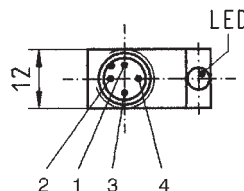
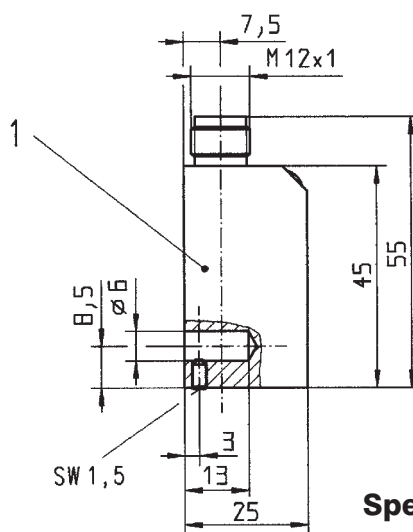
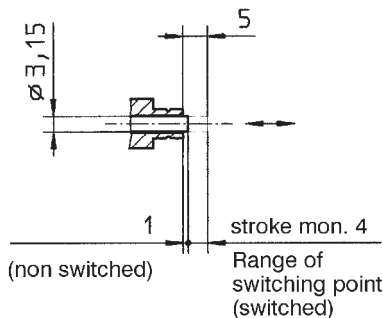
DATA SHEET 66925 - 1311



Dimensions for connection of monitoring switch



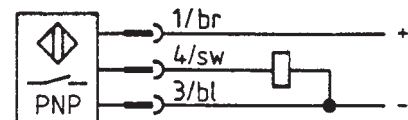
Way of motion indicator for actuation of monitoring switch



Specification

Protective system : ☐ IP 65 in plugged condition
Operating voltage : ☐ 10 V to 30 V DC
Output current : ☐ max. 200 m A
Switching function : ☐ NO contact
Switching frequency : ☐ max. 1000 Hz
Temperature range : ☐ - 25 °C to + 70 °C
Feed lines:
Proof against wrong poling : ☐ yes
Shortcircuit proof : ☐ no

Connection diagram



Example for ordering

Monitoring switch consist of:

1 pcs. Monitoring switch,
Code no. 66925 - 1311

1 pcs. Right angle plug, cable
length 10 m and cable sheathing of
PVC (oil resistant)
Code no. 76928 - 2833

Item.	Pcs.	Designation	Code no.
1	1	Monitoring switch	66925 - 1311
2	1	Cable sheathing of PVC (oil resistant) cable length 2 m ("I")	---
		cable length 5 m ("I")	76928 - 2813
		cable length 10 m ("I")	76928 - 2823
		cable length 2 m ("I")	76928 - 2833
3	1	Right angle plug	76928 - 2843
		cable length 5 m ("I")	76928 - 2853
		cable length 10 m ("I")	76928 - 2863
		Plug (straight)	76928 - 2863